

RECOMMENDED PRACTICE: **ULTRAVIOLET GERMICIDAL** **IRRADIATION (UVGI)**

AN AMERICAN NATIONAL STANDARD

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AN AMERICAN NATIONAL STANDARD**

Publication of this document
has been approved by IES.
Suggestions for revisions
should be directed to IES.

**Prepared for IES
IES Photobiology Committee**



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1.0 Introduction and Scope

1.1 Introduction

The 2014 outbreak of the Ebola virus and the recent global spread of the SARS-CoV-2 virus (and its resulting disease, COVID-19) have renewed interest in the use of ultraviolet irradiation for disinfection. UV radiant energy was first used for disinfecting surfaces in 1877,^{1,2} for water in 1910,³ and for air in 1935.⁴ Ultraviolet germicidal irradiation (UVGI)'s use in recent decades has been largely limited in the U.S. to water treatment facilities, applications where it is hidden (shielded) in heating and air-conditioning ductwork, and use in biological laboratories. UVGI is being used in many countries to control airborne transmission of tuberculosis (TB). In addition, some U.S. healthcare facilities are now using autonomous mobile units ("robots") to add enhanced hygiene to patient rooms in order to reduce hospital-acquired infections. More-widespread use of UVGI is often limited by safety concerns, but these are relatively manageable. UVGI related injuries typically resolve within 24 to 48 hours, and potential increased risk for skin cancer is relatively insignificant and minor compared to a true sunburn⁵ and to potential infection prevention. Most of the public are not aware of its unique value in disinfection of air and contaminated surfaces.⁶ More widespread use of UVGI technology in a wide range of applications should be encouraged in the battle against the COVID-19 pandemic, and this is an urgent matter. At the same time there is a longer-term interest at stake, as UVGI technology can be a vital tool in mitigating and preventing many more current and future viruses. COVID-19 will not likely be the last pandemic.

The term *germicidal UV* (GUV) is sometimes used to refer to ultraviolet radiant energy that can be used to inactivate bacteria, protozoa, mold spores, fungi, or viruses. When the process is applied in a given location, it has generally been referred to as ultraviolet germicidal irradiation (UVGI). Because of the public's concern about *ionizing* radiation (e.g., X-rays and gamma rays), the term *GUV* is sometimes used to avoid needless concerns about a link with that type of radiation, although there are other potential safety concerns (see **Section 5.3**). Another non-technical term is *germicidal light*, although

"light" is usually considered only visible radiation.⁷ Others may refer to the spectral band and term it *germicidal UV-C*.

1.2 Scope

This document addresses the use of germicidal ultraviolet radiation for disinfection of room air and surfaces; potential dangers for humans; potential effects on materials and plants; UVGI technologies; and safety measures and precautions. It does not address photobiological safety with respect to emission or exposure limits; those topics will be addressed in an update to *ANSI/IES RP-27-20, Recommended Practice: Photobiological Safety for Lighting Systems*.⁸

2.0 Definitions and Abbreviations

2.1 Definitions

air changes per hour (ACH): A measure of the clean air volume added to or removed from a space in one hour divided by the volume of the space. Sometimes called air exchange rate.

equivalent air changes per hour (eACH): The number of ACHs that would be needed in order to provide the equivalent inactivation rate of airborne bacteria or viruses using an upper-room UV-C system.

excimer lamp: A discharge source of ultraviolet radiation produced by excited dimer[†] ("excimer") molecules as they transition to their ground state. The wavelength of UV radiation emitted depends on the type of noble gas, which is, depending on the type of source, combined with a halogen gas. Krypton-chloride excimer (chemical symbol: KrCl*) lamps emit primarily at 222 nm, with lesser emissions at longer UV wavelengths. (The asterisk in the symbol indicates that the molecule is an excimer.)

fluence (radiant fluence): The omnidirectional radiant energy externally incident on an elementary sphere about the point, per unit cross-sectional area

[†] A dimer is a molecule or molecular complex consisting of two identical molecules linked together.

of the sphere.⁷ Units: Joules per square meter (J/m²) or millijoules per square centimeter (mJ/cm²).

$$F(\Delta t) = dQ/da = \int_{\Delta t} \int_{4\pi} L \, d\omega \, dt ,$$

where:

a is the cross-sectional area of the sphere, in cm²

L is radiance, W/(cm²·sr)

Q is the radiant energy, J or mJ

t is time, seconds (s)

ω is the solid angle, steradians (sr)

(See also **Sidebar – Fluence vs. Radiant Exposure**.)

fluence rate (radiant fluence rate): The omnidirectional radiant power externally incident on an elementary sphere, per unit cross-sectional area of the sphere.⁷ Also termed *spherical irradiance*. Units: watts per square meter (W/m²).

$$F_t = d\Phi/da = \int_{4\pi} L \, d\omega ,$$

where:

a is the cross-sectional area of the sphere, cm²

L is radiance, W/(cm²·sr)

Q is the radiant energy, J or mJ

Φ is the radiant power, W

ω is the solid angle, steradians (sr)

Hg lamp: In this document: a low-pressure rare gas-mercury (chemical symbol: Hg) lamp.

joule (J): A derived unit of energy; the equivalent of one watt of power radiated or dissipated for one second. 1 watt x 1 second = 1 joule. Or, 1 watt = 1 joule per second.

log inactivation: A term describing a ten-fold reduction in the number of active viruses or microbes; i.e., the number is reduced by 90% for each log step. As an example, if 90% of a sample are inactivated, 10% remain; $0.10 = 10^{-1}$, or $\log(-1)$. Thus, inactivating 90% is a one-log inactivation. Similarly, inactivating 99% (leaving 1%, or 10^{-2}) is a two-log inactivation. [Note: The term “log-kills” is sometimes used. However, this is inaccurate because 1) microbes are not actually killed but prevented from multiplying, and they *eventually* die; and 2) viruses are not alive and thus cannot be killed.]

mycobacteria: Immobile, slow-growing, rod-shaped gram-positive bacteria; includes pathogens known to cause tuberculosis and leprosy.

pulsed xenon lamp: A highly specialized type of gas-discharge lamp; produces a bright white light that closely mimics sunlight, with a spectrum extending into the UV-C range.

radiant exposure: The time-integrated irradiance at a surface. Units: Joules per square meter (J/m²) or millijoules per square centimeter (mJ/cm²).

(See also **Sidebar – Fluence vs. Radiant Exposure**.)

radiant exposure rate: irradiance at a surface. Units: Watts per square meter (W/m²).

SARS-CoV-2: Severe acute respiratory syndrome coronavirus 2019; responsible for COVID-19.

Sidebar – Fluence vs. Radiant Exposure

It is important to note the difference between *fluence* and *radiant exposure*. Both have the same units, J/m² or J/cm². This has confused many scientists, with the result that the terms are often misused, with one substituted for the other. However, the concepts are different and the distinctions are important. The quantities *irradiance* and *radiant exposure* are what instruments measure at *an exposed surface* (and follow Lambert’s cosine law), but *fluence rate* and *fluence* include backscattered radiation and are useful for photochemical calculations *within tissue*, as in photodynamic therapy, in photodermatology or in studies where radiant energy arrives from many directions, such as the exposure of plankton in the sea or aerosolized microorganisms. These quantities are particularly useful in theoretical calculations of “dose” distributions where photochemistry at the molecular level in a scattering medium (e.g., tissue) is enhanced as a result of multiple scattering events in tissue. *Fluence* accounts for this; *radiant exposure* does not. Several other names have been applied to fluence rate, including *spherical irradiance* and *scalar irradiance*.

Threshold Limit Value® (TLV)⁹: The theoretical level of a substance to which a worker can be exposed day after day for a working lifetime without adverse effects. TLV guidelines were developed by the American Conference of Governmental Industrial Hygienists (ACGIH), a professional organization of health protection specialists, for exposure limits for chemical airborne contaminants and physical agents, as well as biological exposure indices (BEI). In 1967 ACGIH provided its first, interim guideline expressed as a spectrally weighted effective UV radiant exposure of $3 \text{ mJ} \cdot \text{cm}^{-2}$ ($30 \text{ J} \cdot \text{m}^{-2}$). The TLV was expressed as an effective photobiological dose, i.e., a time-integrated spectral irradiance (E_{UV}). The unweighted irradiance E_{λ} is spectrally weighted by the UV hazard function $S(\lambda)$ ⁹:

$$E_{\text{UV}} \cdot t = \sum E_{\lambda} \cdot S(\lambda) \cdot t \cdot \Delta\lambda \leq 3 \text{ mJ} \cdot \text{cm}^{-2} \cdot \text{sr}^{-1} \text{ effective}$$

Note 1: $S(\lambda)$ is the action spectrum (i.e., spectral weighting function) for describing the relative UV hazard (most specifically to the eye, but applied to the skin as well) as a function of wavelength between 180 nm and 400 nm. Spectral weighting of a broadband UV spectrum (e.g., from a xenon arc) is accomplished by weighting irradiance as a function of wavelength against $S(\lambda)$ to determine the "effective irradiance" E_{UV} for comparison to the TLV. E_{UV} multiplied by the time-weighted-average exposure duration t should not exceed $3 \text{ mJ}/\text{cm}^2$ (effective) in a typical 8-hour work day. For longer exposure durations (e.g., 24-hour time-weighted average exposure duration for hospital workers or convalescing hospital patients in rooms with upper-room UV lamps), the effective time-weighted average irradiance E_{UV} should not exceed $0.1 \text{ } \mu\text{W}/\text{cm}^2$. (Time-weighted average assessments are discussed in **Section 8.1.2** and **Figure 8-3**.) This function is applied in lamp safety standards from the American National Standards Institute and the Illuminating Engineering Society (ANSI/IES RP-27-20⁷) and the International Electrotechnical Commission (IEC 62471/CIE S009¹⁰), as well as by ACGIH and the International Commission on Non-ionizing Radiation Protection (ICNIRP) for assessing hazards.

Note 2: For the purposes of this document, the lower wavelength limit is set as a practical limit at 200 nm because of radiometric difficulties.

Note 3: The ACGIH TLV limits in 2020 for UV-C exposures are identical to the NCIRP exposure guidelines for the general public. (See also Note 1 associated with this definition.)

Note 4: In the ACGIH process, a new limit is proposed as a "Notice of Intent to Establish" or a "Notice of Intent to Change" a TLV, and the proposal is open for public review and comment for a year.

ultraviolet radiation: Radiant emission within the wavelength range 100 to 400 nanometers (nm).⁷ The subdivisions of the ultraviolet part of the electromagnetic spectrum⁷ (see also **Section 3.0 Types of UV Radiation**):

- *UV-A:* Ultraviolet spectral band 315 to 400 nm
- *UV-B:* Ultraviolet spectral band from 280 to 315 nm
- *UV-C:* Ultraviolet spectral band from 100 to 280 nm
 - *Vacuum UV:* a subset of UV-C, 100 to 180 nm

ultraviolet germicidal irradiation (UVGI): The use of ultraviolet radiant energy to inactivate bacteria, mold spores, fungi, protozoa, or viruses.

upper-room UV-C system: Upper-room UV-C systems include two components: 1) a luminaire with a UV-C source that irradiates the upper portion of a room, with minimal UVGI in the occupied space (generally below 1.8 m (6 ft); and 2) a means to ensure adequate mixing of air between the occupied space and the irradiated zone.

whole-room UV-C system: This type of system irradiates the entire room. These systems are generally designed to provide high-intensity fluence rate in the air and on surfaces. These systems require safety measures if the room is occupied or someone tries to enter a room.

2.2 Abbreviations

ACH: Air changes per hour; used in mechanical ventilation systems (see **Section 2.1**)

ACGIH: American Conference of Governmental Industrial Hygienists

AHU: Air handling units

ANSI: American National Standards Institute

ASHRAE: American Society of Heating, Refrigerating, and Air-Conditioning Engineers

CDC: U.S. Centers for Disease Control and Prevention

CIE: International Commission on Illumination (*Commission Internationale de l'Eclairage*)

DNA: deoxyribonucleic acid

eACH: Equivalent air changes per hour in disinfection produced by UV-C (see **Section 2.1**)

FDA: U.S. Food and Drug Administration

HEPA: High-efficiency particulate air; describes a type of filter or filtering system

Hg: Chemical symbol for mercury

HVAC: Heating, ventilation, and air conditioning

ICNIRP: International Commission on Non-Ionizing Radiation Protection

IEC: International Electrotechnical Commission

KrCl*: symbol for krypton-chloride excimer (the asterisk indicates the excited dimer, or excimer, state)

LED: light-emitting diode

NIOSH: National Institute for Occupational Safety and Health (U.S.)

nm: nanometer; equal to one-billionth of a meter (10^{-9} meter)

PPE: Personal protective equipment

RNA: ribonucleic acid

TLV: threshold limit value (see **Section 2.0 Definitions**)

TWA: Time-weighted average. (*Note:* TLVs applied to UV radiation exposure are typically time-weighted averages; see **Section 8.1.2** for more information on time-weighted averaging.)

UVGI: ultraviolet germicidal irradiation

WHO: World Health Organization

Xe: Chemical symbol for xenon

μm: abbreviation for micrometer

3.0 Types of UV Radiation

The ultraviolet part of the electromagnetic spectrum is generally divided into ultraviolet-A, ultraviolet-B, and ultraviolet-C (UV-A, UV-B, and UV-C).

Wavelengths in the part of the UV-C band from 200 to 280 nanometers (nm) have been shown to be the most effective for disinfection, although longer, less energetic UV wavelengths can also disinfect if applied in much greater doses. UV-C wavelengths comprise photons (particles of light) that are of the highest energy in the optical spectrum[‡] and therefore are the most photochemically active. (See **Figure 3-1**.) It is also important to note that vacuum UV radiation (100 to about 180 nm) is strongly absorbed by air. UV-C radiation, while emitted by the sun, does not reach Earth's surface. In the UV-C range, however, radiation below 200 nm, particularly the 185-nm emission from a low-pressure mercury lamp, can strongly dissociate oxygen molecules to create ozone, which is potentially toxic.[§]

[‡] Optical radiation includes infrared, visible, and ultraviolet.

[§] Ozone has been used as a disinfectant within confined (unoccupied) spaces, but concentrations required are orders of magnitude above applicable limits.

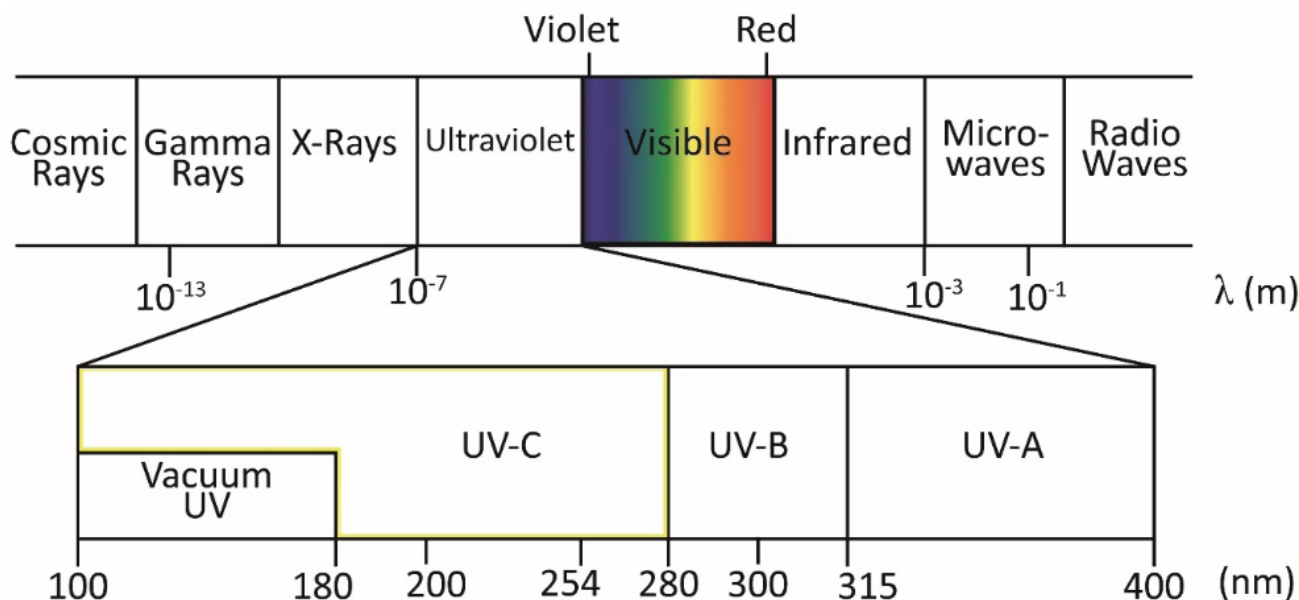


Figure 3-1. The ultraviolet portion of the electromagnetic spectrum. (Copyright Illuminating Engineering Society).

4.0 Types of GUV Equipment and Systems

4.1 GUV Technologies

Although germicidal UV encompasses a continuous range of UV-C wavelengths from at least 207 nm up to 280 nm, in practice predominant wavelengths are determined by the type of lamp or solid-state source, with or without filters, as needed for safety. Source technologies commonly used for disinfection in air or on surfaces include:

- Low-pressure mercury (Hg) lamps, producing predominantly 254 nm UV
- Excimer lamps, particularly KrCl* (see **Section 2.1 Definitions**), producing predominantly 222 nm UV and frequently used with filters to eliminate longer wavelengths
- LEDs, variable in spectral and optical output in the UV-C range and requiring no filter to block longer UV-B wavelengths

From an electrical efficiency perspective, low-pressure mercury lamps are most efficient and have a 6,000- to 8,000-hour life (about one year of continuous use). Much of that efficiency is lost when Hg fixtures are louvered to prevent occupant overexposure to UV radiation.

Currently, LEDs are least efficient, and the efficiency goes down dramatically as the wavelength is reduced. The life of LED lamps is not known well at this time. KrCl* lamps are intermediate in efficiency and can potentially be used directly in occupied rooms. They generally have a 3,000-hour lamp life, though some may be longer. More-complex sources, such as pulsed xenon arc lamps, are used in particular applications where large amounts of UV are needed.

Medium-pressure mercury lamps resemble high-pressure mercury lamps—i.e., they typically have much higher power—and use a clear or doped quartz envelope, depending on the application. While the efficiency of the 253.7-nm emission is significantly reduced and broadened due to reabsorption of the radiant energy, it is nevertheless an important source, as it can be operated at much higher power per unit length compared to low-pressure mercury lamps. Medium-pressure mercury lamps have found some application as a disinfection source in HVAC systems, but their primary use is in water disinfection or purification, as the water provides greater heat removal from the source surface compared to air—a significant advantage in using lamps with very high power.

All of the above technologies—continuously emitting or pulsed—are comparably effective for disinfection

and depend only on wavelength, not type of source. Pulsed sources may be needed if rapid disinfection is required.¹¹

4.1.1 Low-Pressure Mercury (Hg) Lamps. Currently, the most practical and most efficient method of generating germicidal radiant energy is through the use of low-pressure mercury lamps. These lamps are electrically the same as fluorescent lamps (hot-cathode starting) of corresponding sizes and wattages and require essentially the same auxiliary equipment (ballasts). They differ physically from fluorescent lamps in that they contain no phosphor and are constructed with a special kind of glass that transmits most of the UV energy generated by the Hg discharge. **Figure 4-1** shows examples of low-pressure Hg germicidal lamps. The effective life for germicidal low-pressure Hg lamp is usually significantly shorter than that of fluorescent lamps, as maintaining the transmission of the UV radiation through the glass over life is more difficult than for fluorescent lamps.

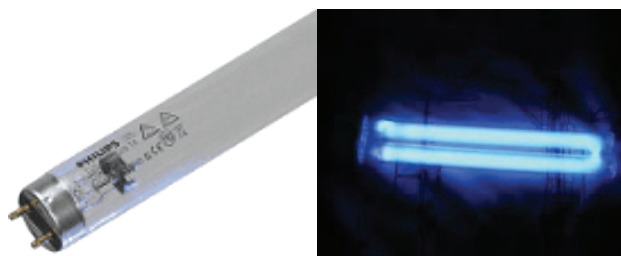


Figure 4-1. Examples of germicidal low-pressure Hg lamps. (Note the added fluorescence and scattered output in the photo on the right.)

Low-pressure mercury germicidal lamps with cold-cathode electrodes are also available in various sizes, usually for shorter, smaller-diameter lamps that are used in wands and some air purifiers. Their operating characteristics are similar to those of hot-cathode lamps, but their starting mechanisms are different.

Approximately 45% of the input power from a low-pressure Hg lamp is emitted from the discharge at a wavelength of 253.7 nm (usually rounded to 254 nm), in the middle of the useful UV-C band. The

second major emission line is at 184.9 nm (usually referred to as 185 nm), but this emission is absorbed by the glass. If emitted through the glass, as it is with pure quartz, it would create ozone at levels far above the safety limit. Other mercury lines in the UV-B and UV-A regions are present at much lower emitted-power levels (about 7% total), half of which is due to the broad baseline emission peaking at about 300 nm. While the total UV-B emission is about 3.5% of the 254-nm emission, recent models¹² show that UV-B penetrates significantly deeper into the skin than the 254-nm emission; therefore, while the emission of UV-B is small, it may be more of a safety concern in terms of erythema or skin cancer than the 253.7-nm emission. **Figure 4-2** shows measured spectral irradiance from a germicidal low-pressure Hg lamp. The strength of the emission at the various wavelengths is plotted logarithmically. The special glass transmits about 80% of the 254-nm radiation; thus, nearly half the input power from the lamp is available for UV disinfection (see **Sections 5.0** through **5.2**). (Note the continuous energy peaking at about 300 nm due to *bremssstrahlung*, i.e., photon emission due to collisional deceleration of electrons in an electric field.)

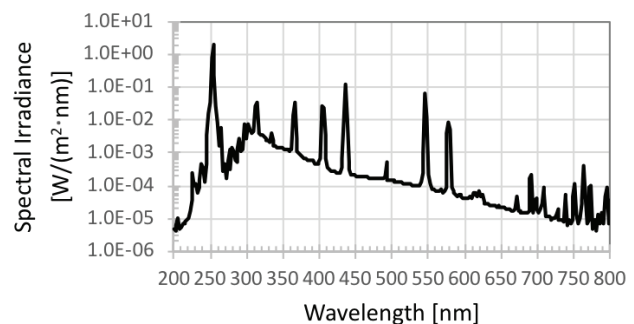


Figure 4-2. Typical UV and visible emission spectrum for a germicidal low-pressure Hg lamp (GE G30T8) expressed on a logarithmic scale. (Graphic courtesy of Rolf Bergman)

4.1.2 Medium-Pressure Mercury Lamps. Medium-pressure mercury lamps are almost exclusively used in water purification. Such lamps resemble high-pressure mercury lamps—i.e., typically have much higher power—and use a clear or doped quartz envelope, depending on the application. While the

efficiency of the 253.7-nm emission is significantly reduced and broadened due to reabsorption of the radiant energy, it is nevertheless an important source, as it can be operated at much higher power per unit length than low-pressure mercury lamps. When used in water disinfection, the water also provides higher heat removal from the glass surface than air can, which allows the use of much higher power density.

4.1.3 Xenon Lamps. Xenon (Xe) arc lamps, which can be short-arc lamps or long-arc lamps, generate substantial UV radiation. Such lamps are made with quartz envelopes, usually containing a doping agent to prevent emission below 200 nm, as these shorter wavelengths can generate ozone. Further, particularly for short-arc lamps, a pulsed current input generates much greater emission. **Figure 4-3** shows that the spectrum of the pulsed lamp is also broadened so that it becomes a nearly continuous emission, particularly in the UV-C region. Thus, xenon lamps can simulate the emission spectrum of sunlight. High-pressure xenon lamps need to be encased because the high pressure can cause the lamp to rupture in operation.

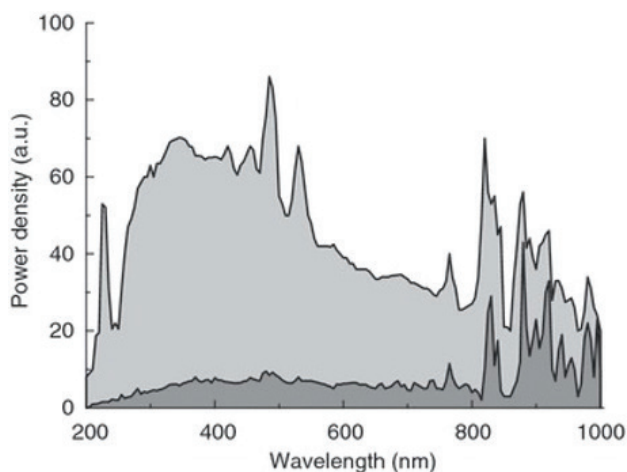


Figure 4-3. Difference in spectral characteristics between continuous (dark gray) and pulsed (light gray) modes of a xenon lamp.

4.1.4 Excimer Lamps. Lamps using excimers—photon emission from semi-stable dimers, e.g., XeXe* (as with KrCl*, the asterisk indicates the excited dimer, or excimer, state)—have been around

for some time. Some rare-gas-halogen excimer (e.g., krypton-chloride, KrCl*) lamps have been shown to produce significant emission in the UV-C region between 200 and 230 nm. The advantage of sources such as KrCl*, which has a strong emission peak at 222 nm, is that the deactivation rate of some bacteria and viruses appears to be about the same as, or possibly higher than, that achieved with 260 to 280 nm, while the effect of the shorter-wavelength UV-C emission on human skin and eyes (see **Section 5.3**) is much reduced compared even to the 253.7-nm mercury emission.¹³ However, small but significant emission levels at around 259 nm, on the order of 20 to 100 times smaller than the main 222-nm emission, may be of biological concern.^{14,15} There is a concerted effort to reduce the longer-wavelength emission through use of special filters. At this time, excimer sources are mainly used in specific applications, but their usage in potentially occupied environments is rapidly growing. However, the potential benefit of using these sources in places where humans are present will likely spur their commercial development.

UV emission below 240 nm can produce ozone at very low levels, which increases with shorter wavelengths, particularly below 200 nm. The U.S. Occupational Safety and Health Administration (OSHA), based on ACGIH guidelines,⁹ has set permissible exposure levels (PEL) that allow a maximum time-weighted average (TWA) level for ozone exposure of 8 hours at 0.1 parts per million (ppm) for “light” work, and 8 hours at 0.05 ppm for “heavy” work.

Ozone concentration in rooms is an important design consideration when using 222-nm lamps. Depending on design and power, KrCl* lamps can lead to ozone production of potential safety concern that may require mitigation. This has to be taken into account and measured when designing and deploying fixtures with KrCl* lamps. Room size and adequate ventilation to disperse ozone when using excimer lamps have to be considered. It is important that users read manufacturers’ instructions in this regard.

4.1.5 LEDs. There are several companies producing LEDs that emit in specific wavelengths of the UV-C region. Such LEDs are being tested for upper-room disinfection, occupied-room direct disinfection, and surface disinfection. This is a promising area of research, and such LEDs should be considered as further data becomes known. (Refer also to the relevant tables in **Section 9**.) Further, the fact that LEDs are directional emitters provides advantages for the use UV-C LEDs in upper-room UVGI systems, as the need for complex reflectors and baffling louvers are not required. Relatively simple lenses used with the LEDs are all that are required to produce luminaires with collimated emission. Thus, while the efficacy of UV-C LEDs in the region of 265 to 275 nm is currently about an order of magnitude below that of the low-pressure mercury lamps, the actual difference in delivered irradiance may not be much different compared to a low-pressure Hg system with louvers, since the light can be much more efficiently directed. In addition, large increases in efficacy and decreases in cost are expected over the next 10 years. As yet, complete upper-air GUV LED systems are not on the market, but with the emergence of the SARS-CoV-2 virus, these may become a reality in the very near future. Another characteristic of LEDs is that they can be rapidly pulsed, if needed, since the radiation emitted from LEDs responds to the controlling electrical current within microseconds.

A disadvantage of the LEDs operating in the region of 265 to 275 nm is that their emission is near the UV-B region, and thus the possibility of erythema and potential for skin cancer are greater than with the same dose produced by low-pressure mercury and excimer lamps. This is of less concern for LEDs emitting below 260 nm.⁵ Another disadvantage of LEDs is that heat dissipation is needed. Thermal management plays an important role in LED-based luminaires and is a critical aspect of the design process. The UV emission, life, and reliability are all negatively affected with increasing temperature.

LED developers have made rapid improvements in sources that emit at lower wavelengths in the UV-C region. These developers continue to innovate

solutions to obstacles with current materials and processes, building on the lessons learned from white-light LEDs to focus on the unique aspects of UV-C LEDs for longer lifetimes, higher efficiencies, and greater availability.

Recently, a product was introduced using LEDs emitting at 255 nm, a comparable wavelength to the main emission line of low-pressure Hg (253.7 nm). While the photon efficiency of these LEDs is currently on the order of 0.1%, they may prove important even at such low efficiency, due to the ease of focusing the emission as well as divisibility of sources for installations. **Figure 4-4** shows one view of the state of development of UV-C LEDs. It should be noted from this figure that quantum efficiencies decrease with wavelength and that no devices below 255 nm are shown. Thus, development of LEDs with emission comparable to that of the KrCl* lamp appears to be not a viable solution within the near future.

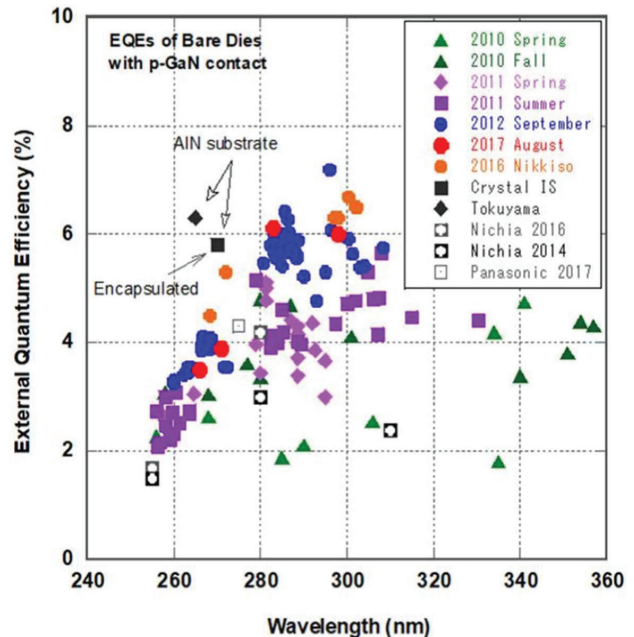


Figure 4-4. UV-C LEDs: External quantum efficiency as a function of wavelength.¹⁶ (Abbreviations in figure: EQE = external quantum efficiency; p-GaN = p junction of gallium nitride; AlN = aluminum nitride)

4.2 In-Duct Systems – Surface and Air

Coil and drain pan surface irradiation systems: These systems are installed within HVAC air-handling units

and duct runs. Dark and damp conditions within these systems can promote the growth of bacteria, mold, and other biological entities. As air-conditioning equipment ages, its ability to maintain designed air temperatures and humidity levels decline. Predominantly, the culprit is reduced coil-heat-transfer efficiency due to biofilms on coil fins, which reduces the ability of the air handling unit's cooling coil to remove heat and water from the air. UV-C lamps can be installed to target problematic components such as cooling coils or condensate pans, or applied to give a broad distribution of germicidal energy over an entire enclosure (e.g., a mixing box or plenum) that might have microbial activity. Systems and equipment may be verified for GUV output through *ANSI/ASHRAE Standard 185.2-2014, Method of Testing Ultraviolet Lamps for Use in HVAC&R Units or Air Ducts to Inactivate Microorganisms on Irradiated Surfaces*.¹⁷

HVAC airstream-disinfection systems: UV-C air disinfection systems are installed in HVAC ducts, in air-handling units or air distribution systems to inactivate microorganisms and viruses and disinfect moving airstreams “on-the-fly.” UV-C exposure and the resulting dosage are determined by the amount of germicidal energy absorbed by a pathogen over a specific length of time. In-duct air disinfection systems should be designed to have the desired single-pass inactivation level under an application's worst-case conditions of air temperature and velocity in the irradiated zone. Airborne inactivation rates over 99% on a first-pass basis¹⁸ have been modeled, and as air is recirculated, concentrations are further reduced by each subsequent pass (mass balance). Systems and equipment may be verified for GUV output through *ANSI/ASHRAE Standard 185.1-2015, Method of Testing UV-C Lights for Use in Air-Handling Units or Air Ducts to Inactivate Airborne Microorganisms*.¹⁹

4.3 Upper-Room Systems

Upper-room UVGI is a primary means of safe air disinfection that is possible in rooms with high ceilings. In this method, specially designed and installed UV-C fixtures that irradiate only the air above 2.1 meters (7 feet) constantly disinfect the upper air volume. This is most effective when there is constantly mixed air by fans and HVAC ventilation, but even without strong ventilation or fans, air constantly mixes by movements and normal convective currents.

In practice, upper-room UVGI should augment building ventilation for disinfection and should not be a standalone method of disinfection. For surface decontamination, especially in healthcare, manual cleaning of high-touch surfaces is necessary prior to using UVGI as a method for terminal cleaning. This is because there may be buildup on surfaces of bioburden that is not well penetrated by UV-C radiation. Similarly, it is important that lamps also be cleaned regularly.

4.4 Mobile Systems, Including Robots

Hospitals and healthcare facilities have rooms that can be closed off to individuals for a length of time. So-called “UV-C robots” have been used to move around a room to disinfect surfaces with UV-C in all directions. The UV-C radiant energy is normally emitted by long, vertical mercury lamps or pulsed xenon lamps. It is a challenge to estimate dose, but very intense emission can cover much of the room in a relatively short time. Further, by moving autonomously around the unoccupied workspace, it can expose surfaces that would not be easily reached by fixed UVGI equipment installations. If good air movement is present, most air will be disinfected as well, and the dose requirements noted in **Section 6.1 Air Disinfection** could be applied. Thick buildup of residues on surfaces may absorb the UV-C photons before they reach the active virus or bacterium; therefore, it is important to include surface cleaning as part of the infection prevention and control process. In addition, regular cleaning of the GUV luminaires (including lamps) will ensure maximum performance. As with all GUV systems, these systems should be considered as an effective *adjunct* to standard infection-control cleaning guidance. These mobile units should be used *after* patient departure for cleaning of patient rooms and bathrooms.^{20,21}

4.5 Handheld Devices

Hand-held, compact GUV products (see **Figure 4-5**) have been marketed for more than a decade for disinfecting small objects such as cell phones. Most of these emit less than 2 mW/cm² of 254-nm UV-C at contact, meaning that the wand has to be held close to the surface for several seconds for an effective multi-log unit disinfection. Waving it over an object such as a postcard for one second will not provide reliable disinfection. These products typically employ a safety

switch that senses when the emission is not directed downward (away from the eyes) and shuts off if turned upward. Even if safely used, these might provide a false impression of effective disinfection. It is critical that these devices be used with extreme caution, as there can be UV exposure even if the device is facing away from the user, because of material reflections. The reflectivity of surfaces that are irradiated or within the line of sight of the device needs to be considered for safe use. While most surfaces do not reflect well in the UV-C region,²² this is not true for all materials.



Figure 4-5. Small UV-C mercury lamps and 270-nm LEDs used in hand-held wands are available on the internet (middle and bottom). Small air-space disinfectant units (top) are also sold. (Image courtesy of David Sliney)

5.0 Effects of Ultraviolet Radiation

Ultraviolet radiation has a myriad of effects, both beneficial and harmful, on animals (including humans), plants, and non-living substances such as paints and fabrics. Some of these effects are harnessed for their usefulness, while adverse effects would require protective action. One of the most important effects, from the standpoint of this document, is the use of UV radiation to disable pathogens.

5.1 How UVGI Disables Bacteria and Viruses

Individual, energetic UV-C photons photochemically interact with and damage the RNA and DNA molecules in a virus or bacterium, thus preventing replication and rendering these microbes non-infectious. This includes SARS-CoV-2 if the virus is directly irradiated by UV-C at

an effective dose level. This all happens at a microscopic level. Viruses are much less than one micrometer (μm , one-millionth of a meter) in size, and bacteria are typically 0.5 to 5 μm . Thus, the largest viruses are about the same size as the smallest bacteria, but for the most part, viruses are much smaller than bacteria.

UV-B (280 to 315 nm), particularly the shorter wavelengths near 300 nm and below, can be relatively effective as a germicidal source, but in accidental exposures there is a significantly higher risk for severe erythema (akin to sunburn) and even delayed effects for both skin and eyes, because UV-B penetrates the skin more deeply. (See **Section 5.3 Dangers to Humans.**)

UV-A and longer (visible) wavelengths do not have germicidally effective emission wavelengths to inactivate viruses. Their relative disinfection capability in other microbes is minimal—on the order of 1,000 times less effective in terms of fluence rate than the UV-C emitted by low-pressure mercury germicidal lamps. There have been only very special applications of wavelengths in the UV-A and violet (e.g., 405 nm) range²³; these require very high doses that are seldom practical in an occupied environment, and therefore meaningful infection control has not been shown.

Light-emitting diodes (LEDs) have been available for some time in the UV-A region. The advantages of UV-A or visible-light LEDs would be that they can easily be incorporated into LED-based luminaires, and there might be no need for protective gear. However, as mentioned above, the efficacy of violet or UV-A energy that is not harmful to the skin or eyes is minimal. (See **Section 5.3 Dangers to Humans.**)

Because of its ultraviolet content, direct sunlight has a germicidal effect, particularly in the late spring and early summer when the sun is high in the sky and the UV index is high. At a UV Index of 10, the duration required to achieve at least a three-log inactivation of bacteria (99.9% inactivated; see **Section 5.2 Ultraviolet Disinfection Efficacy**) is estimated as less than one hour.²⁴ (Note: Architectural glass is opaque to ultraviolet at wavelengths below approximately 350 nm, while acrylic plastic has a cutoff of approximately 390 nm. This would prevent “natural” disinfection of interior sunlit spaces.)

5.2 Ultraviolet Disinfection Efficacy

The efficacy of an installed system will depend on the specific microbial pathogen(s) being targeted, the characteristics of the system, and the characteristics of the installation. The extent of inactivation will be governed by the dose of UV radiation delivered to the target microbes.

The most fundamental concept in photobiology is the action spectrum (or relative response) for a given effect. Although there is a standardized germicidal action spectrum, it is based on inactivation of *E. coli* bacteria (see **Table 5-1** and **Figure 5-1**). It is important to note that action spectra for spores, other bacteria, protozoa, fungi, and viruses can vary from the standardized action spectrum.^{13,25} This standardized action spectrum extends from 235 nm to 313 nm and peaks at approximately 265 nm. A wavelength of 254 nm has a relative efficacy of 0.85; by contrast, 313 nm in the UV-B has a relative efficacy of only 0.01.

Table 5-1. Bactericidal Efficiency (for a Common strain of *E. coli*) of UV Radiation at Mercury Emission Lines

Wavelength (nm)	Relative Efficiency
235.3	0.35
244.6	0.58
248.2	0.70
253.7	0.85
257.6	0.94
265.0 ⁱ	1.00
265.4	0.99
267.5	0.98
270.0	0.95
275.3	0.81
280.4	0.68
285.7	0.55
289.4	0.46
292.5	0.38
296.7	0.27
302.2	0.13
313.0	0.01

ⁱ Interpolated

Source: IESNA Lighting Handbook, 9th ed.

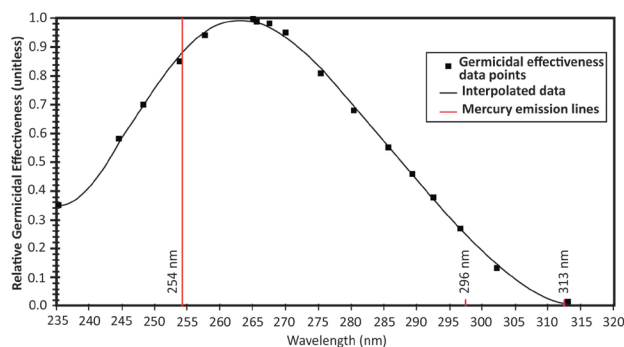


Figure 5-1. Graphic display of the data from Table 5-1 (specific to a common strain of *E. coli*).

Germicidal effectiveness is proportional to the exposure dose (typically measured in millijoules per square centimeter, mJ/cm², or joules per square meter, J/m²), which is the product of the dose-rate (germicidal action spectrum-weighted effective irradiance, typically in effective mW/cm² or effective W/m²) and time (from 1 microsecond [μs] to several hours). A nonlinear relationship exists between UV exposure and germicidal efficacy. For example, if a certain UV exposure inactivates 90% of a bacterial population (referred to as a one-log inactivation), at least twice the exposure would be required to inactivate 90% of the residual 10%, for an overall germicidal efficacy of 99% (a two-log inactivation). Likewise, a decrease of at least 50% in dose or exposure time would decrease germicidal efficacy from 99% to 90%.

Generally, a desired UVGI dose of interest is one that will accomplish a 3- or 4-log inactivation, corresponding to 99.9% or 99.99% inactivation, respectively, and set with a duration (typically per hour) so that an eACH can be readily calculated. However, in practice, achieving a 2-log inactivation (99% inactivation) is frequently accepted.

Germicidal UV irradiation is effective for inactivation of a wide range of microbial pathogens. For many microbes and viruses, the germicidal effectiveness of UV irradiation improves as relative humidity (RH) decreases. This behavior has been attributed to changes in the structural form (and thus function) of nucleic acids, which may alter the fundamental photochemistry of these critical biomolecules. Some studies^{26,27,28,29,30} have suggested that desiccation as a form of general

stress makes microbes and viruses more susceptible to the effects of UV-C exposure. This effect can vary significantly, depending on the pathogen.

Many simple microbial pathogens, including many viruses, display first-order inactivation responses. This results in a linear relationship between the log of target pathogen inactivation and the UV-C dose delivered to the pathogen population:

$$\ln(N/N_0) = -kD,$$

where:

N = concentration of viable (infective) pathogen after exposure

N_0 = concentration of viable (infective) pathogen before exposure

k = inactivation rate constant (cm^2/mJ)

D = applied UV dose (mJ/cm^2)

In this regard, k is a measure of the sensitivity of a target pathogen to UV exposure. Alternatively, the sensitivity of a pathogen to UV exposure may be the dose needed to achieve 1 $\log_{10}$ unit of inactivation, sometimes referred to as D_{10} .[¶] **Table 5-2** provides a summary of reported values of k and their corresponding D_{10} values for several single-stranded RNA (ssRNA) viruses exposed to UV-C at 254 nm.

[¶] Sometimes D_{90} is seen instead of D_{10} . D_{90} refers to 90% inactivated, while D_{10} refers to 10% remaining.

Table 5-2. Reported or Estimated Values of k and D_{10} for Several ssRNA Viruses in Air

Virus	k (cm^2/mJ)	D_{10} (mJ/cm^2)
MS2 bacteriophage	5.4 – 6.8	0.34 – 0.42
Porcine reproductive and respiratory syndrome virus (PRRSV)	5.8	0.40
Influenza A	2.1	1.1
SARS-CoV-2 ⁱⁱ	3.3	0.69
Severe acute respiratory syndrome virus, 2003 (SARS, SARS-CoV) ⁱⁱ	3.3	0.69
Methicillin resistant staphylococcus aureus (MERS) ⁱⁱ	3.4	0.69
Mouse hepatitis virus (MHV)	3.4	0.67
Equine torovirus (EToV)	3.2	0.71
Sindbis virus (SINV)	0.52	4.4
Venezuelan equine encephalitis virus (VEEV)	0.43	5.3
Infectious salmon anaemia virus (ISFV)	1.4	1.7
Newcastle disease virus (NDV)	0.74 – 0.92	2.5 – 3.1
(Measles virus (MeV)	0.92 – 1.2	2.0 – 2.5
Melbourne H1N1 influenza A virus	1.0	2.3
NIB-4 H3N2-3 influenza A virus	0.92	2.5
NIB-6 H1N1 influenza A virus	1.0	2.2
Isavirus (ISAV)	2.1	1.1
Rabies virus (RABV)	2.3	1.0

ⁱⁱ These values were calculated based on nucleic acid size and structure.

Source: All values are based on exposure to UV-C radiation at a wavelength of 254 nm. Reported values are from Hadi et al.³¹ (2020), and Sagripanti and Lytle (2020).³²

As mentioned above, inactivation responses will also vary with the wavelength(s) emitted by the UV source (see, for example, **Table 5-1**). The vast majority of reported inactivation data are based on exposure at 254 nm, which is the characteristic wavelength of a low-pressure Hg lamp (refer to **Section 4.1.1**). However, alternative sources of germicidal UV radiation have been developed and are being applied for disinfection of air and surfaces. Prominent examples include UV-emitting LEDs (refer to **Section 4.1.5**) and excimer lamps (refer to **Section 4.1.4**). The output characteristics of these sources depend on the material composition of the LED or excimer; most of these sources emit across a relatively narrow wavelength band. Because of this, it is possible to select a wavelength range to use in these applications. System designs should be based on inactivation constants (or values of D_{10}) that correspond with the output spectrum of the UV sources used to develop these systems.

Systems that are used to disinfect air and surfaces should deliver a distribution of UV doses within the irradiated space and to the air that circulates through a room. This dose distribution, together with the kinetics of inactivation, will govern overall system efficacy. The dose distribution delivered by a UVGI system will depend on the spatial distribution of radiant energy within the irradiated room, which in turn is governed by the output characteristics and locations of the UV sources, as well as the geometry and optical characteristics of surfaces in the room. For disinfection of air, the dose distribution will also be influenced by mixing behavior and general fluid mechanics, which in turn are governed by HVAC system design and the geometry of the irradiated room.

UVGI at 222 nm is relatively new and appears to be highly germicidal, but it penetrates surfaces so little that it appears to be safe³³ for direct human skin exposure at doses well above the current 8-hour TLV of 23 mJ/cm², and the risk for eye exposure is lower if the dosage is spread over several hours. This same limited surface penetration may thus limit its utility for surface sterilization. The human exposure limit for UV radiation between 265 nm and 270 nm is half that of 254-nm UV, but ultraviolet exposure in this range should also be somewhat more germicidal (refer to **Table 5.1**, for example), requiring less in the upper room. Normally,

the values in **Table 5.1** are not applied when designing a GUV installation, because of the wide variability of pathogen sensitivities. Designing for TB bacteria is frequently an aiming point.

To summarize, each technology has its advantages. The main advantage of UV LEDs (see **Section 4.1.5 LEDs**) is greater design flexibility compared to light sources of other types. Low-pressure Hg lamps are the workhorse of air and surface disinfection, but humans should not be exposed directly, due to eye and skin irritation. Therefore, 254-nm UV-C is used for room air disinfection indirectly (emitted above people's heads), with good vertical air mixing to ensure air disinfection of the entire room (refer to **Section 4.3 Upper-Room Systems** and **Section 6.1 Air Disinfection**). Low-pressure mercury lamps (254 nm) are also used in unoccupied rooms for surface disinfection (**Sections 4.4** and **6.2**), in ventilation ducts (**Sections 4.2** and **6.1**), and in portable room air cleaners (**Sections 4.4** and **4.5**).

5.3 Dangers to Humans

UV germicidal lamp emissions can pose a workplace safety and health hazard to the eyes and skin if the lamps are improperly used or installed. However, these lamps can be used safely if workers follow appropriate safety precautions. Upper-room UVGI has been safely used for preventing airborne transmission for at least 70 years.³⁴ A great deal is known about the human skin exposure limits of 254-nm UV (UV-C) irradiation. Compared to the UV-A and UV-B in sunlight, UV-C is almost entirely absorbed by the outer dead layer (stratum corneum) and outer skin (outer epidermis), with very limited penetration to the deeper cellular layers of skin where new cells are constantly created. A study³⁵ of continuous monitoring of well-trained healthcare workers and convalescing patients in a hospital with a properly installed upper-room GUV installation recorded UV exposures no more than one-third of the 8-hour ACGIH dose limit and reported no overexposures over a 16-year period,³⁶ thus showing conservative initial assessment levels.

For comparison, the current daily safety limit of 254-nm UV-C for 8 hours is 6.0 millijoules per square centimeter (mJ/cm²), whereas less than ten minutes of summer sun exposure at a UV Index of 10 can deliver the equivalent

limiting daily safety dose⁹ because of its much more-penetrating UV-A and UV-B. Because it has no outer dead protective layer, the human eye is the organ most susceptible to acute sunlight or UV-C exposure. Exceeding the threshold level value (TLV) in the lower room can result in painful irritation of the cornea similar to overexposure on a sunny day, especially from sun reflected from water or snow. The damage is painful but transitory, with corneal shedding of surface cells and replacement in a day or two.³⁷ When the UV-C source is overhead, the eyes receive very little exposure during normal activities; this is demonstrated in sunlight when the sun is overhead—there is hazardous exposure of the skin but not the eyes. There are no known long-term consequences from an accidental UV-C overexposure.⁵ Most eye injuries result from workers on ladders cleaning fixtures or working in the upper room without first turning off the fixtures.³⁷ In some cases, the exposed persons did not fully appreciate the UV-C lamps' potential hazard, or they were unaware that a hazard was present.³⁸ For this reason, only trained maintenance workers should be working in the upper room or replacing in-duct UV lamps. Eye injuries have resulted from insufficient training or improper installation—e.g., workers mistakenly installing an upper-room UV-C fixture upside-down after lamp replacement.³⁸

It is essential that individuals be protected to prevent UV hazards to the eyes and skin as elaborated in **Section 8 Safety Measures and Restrictions**. For general consumers, the International Commission on Illumination warns against using UV-C to intentionally disinfect the hands or any other skin.³⁹ GUV systems shall be designed and installed to ensure that individuals within occupied areas are not exposed above the recommended ACGIH limits.⁹ Needless exposure should be avoided. (Refer to **Section 8** for additional information.)

It should also be noted that although ozone generation is not an optical hazard, it is important that it be assessed after installation of lamps emitting substantially below 240 nm, unless the lamps are certified as "low-ozone producing lamps."

5.3.1 Source-Specific Hazards. UVGI technologies are described in **Section 4.1**. The source-specific

risks to humans vary with the source type. Specific hazards, listed here by UVGI source, include:

Low-pressure and medium-pressure mercury (Hg) lamps: These lamps emit UV energy (low-pressure Hg primarily at 254 nm; medium-pressure Hg in a broad spectrum across the UV-C range) that poses a hazard to the cornea and skin (see **Section 5.3 Dangers to Humans**).

LEDs: UV-C LEDs typically emit between 260 and 280 nm (depending on the type of LED), which poses a hazard to the cornea and the skin (see **Section 5.3 Dangers to Humans**).

Krypton-chloride (KrCl*) excimer lamps: These lamps emit primarily at 222 nm, with lesser emissions from longer UV wavelengths, and may pose a hazard to the cornea and skin if the dose is above the ACGIH limits.⁹ Some KrCl* lamps use narrowband filters, so that wavelengths longer than 222 nm are greatly reduced.^{15,33}

Pulsed xenon arc lamps: These lamps emit UV (in a broad spectrum across the UV-C range) and visible radiant energy that can pose a hazard to the retina, cornea, and skin. Some pulsed xenon arc lamps are filtered so that only the UV energy for disinfection is emitted. Xenon arc lamps can also pose additional safety hazards if they are not maintained properly. These lamps are generally used only in industry, to sterilize food and pharmaceutical containers, for example, but also have been used in robots for hospital room disinfection. Maintenance and service should only be performed by authorized persons.

5.3.2 The Risk of Skin Cancer. In contrast with longer (UV-B) wavelengths, UV-C is absorbed principally in the superficial layers of the skin and eye, with relatively little penetrating to the reproductive cells of the epidermis. In addition, the erythema caused by acute overexposure to UV-C tends to be mild and transitory, with little evidence of subsequent inflammation, blistering, hyperplasia, or induced melanization (i.e., UV-C does not cause tanning). Those latter effects are,

however, associated with UV-B and UV-A exposures. Unlike UV-C, UV-A and UV-B radiation can reach the germinative layer (the basal layer of the epidermis) with increased risk for skin cancer. Low-pressure Hg and KrCl* lamps emit small amounts of UV-B as well as UV-C radiation (e.g., the Hg lamp emission lines at 297, 303, and 313 nm), which theoretically could contribute to delayed effects.⁵ However, UV-C wavelengths are not considered a significant source of increased long-term risk, especially those less than 230 nm.^{14,15,40} However, even though GUV emissions can pose a theoretical delayed hazard, incidental GUV exposures within the ACGIH TLVs would not significantly increase one's lifetime risk for age-related pathologies or skin cancer when compared to the much greater lifetime risk from daily exposure to the UV-A and UV-B radiant energy in sunlight.^{5,41} It should be noted that significant amounts of UV-B and longer wavelengths are emitted from pulsed xenon lamps and therefore pose a much greater risk for delayed effects if not mitigated.

5.4 Effects on Paints, Materials, Plants

UV rays in general can degrade paint, cause plastics to yellow, cause surface degradation, and destroy air filters, depending on their materials (thus, UV-C irradiation of respirators for reuse should only be used as a last resort, such as when there is a shortage of personal protective equipment). Furthermore, shorter-wavelength UV photons have higher energy than longer-wavelength UV photons, and may have an accelerated aging effect on materials and paints. The effect depends upon the color and type of the material; for example, white and transparent plastics are prone to yellowing, and red pigments tend to fade more than blue pigments. UV-C has less dramatic effects than UV-B or sunlight, as it only penetrates the surface of the material but not the structural layers. Wall-hung artwork (e.g., paintings, tapestries) should be protected from direct or indirect exposure to UV-C radiation.

ASHRAE research of materials exposed to continuous UV-C radiation at 254 nm inside air-handling units showed degradation of filters and wire coverings.^{42,43}

UV-C radiation may damage plants; therefore, plants should not be placed in the disinfection zone in upper-room applications or in whole-room UV-C applications.

6.0 Design and Use of UVGI Systems for Disinfection Effectiveness

Commonly used germicidal lamps generate predominantly 254-nm UV radiant energy, which is close to the peak germicidal wavelengths of 265 to 270 nm—both in the UV-C range, compared to the longer-wavelength ultraviolet (UV-A and UV-B) in sunlight. Bacteria and viruses vary somewhat in UV susceptibility, with outdoor environmental organisms, fungal spores, and mycobacteria being relatively harder to kill than more rapidly replicating and non-environmental microbes and most bacteria. But even fungi are effectively killed with high-dose UV, which is used, for example, to treat fungal contamination of air conditioning systems. UVGI can be most effectively used to disinfect air in the upper room, where ceiling height permits, but can also be used in ventilation ducts and room air cleaners, as noted.

Some hospitals have used portable UVGI fixtures to disinfect air and surfaces in unoccupied, locked rooms as a supplemental control measure to reduce the spread of healthcare associated infections.⁴⁴ However, well controlled studies of efficacy are very difficult to conduct and therefore lacking. Medical treatment facilities are using UVGI in four primary ways: 1) upper-room UVGI fixtures with air mixing, for controlling airborne pathogens in an occupied space; 2) whole-room UVGI fixtures (including mobile UVGI units), to disinfect high-touch surfaces; 3) UVGI in HVAC air handling units, to treat recirculated air and to reduce mold growth on cooling coils; and 4) room air purifiers incorporating UVGI. Autonomous (“robot”) systems have been used in some U.S. hospitals and were used in the People’s Republic of China in response to COVID-19.⁴⁵ In fighting an epidemic or a pandemic, as in fighting a war, a single weapon is never used; rather, multiple weapons are exploited.⁴⁶

6.1 Air Disinfection

Upper-room UVGI disinfects large volumes of room air (above occupants' heads; see **Figure 6-1**) at once, resulting in high equivalent air changes per hour (eACH) in terms of air disinfection only—UVGI does not dilute odors or CO₂, the main functions of building ventilation. Odor control and CO₂ removal are accomplished by relatively low levels of ventilation (1 to 2 ACH), but air disinfection requires much higher rates of ventilation (6 to 12 ACH), or the equivalent produced by upper-room UVGI. Two hospital controlled studies^{47,48} have shown upper-room UVGI to be about 80% reduction in spread of tuberculosis (TB) in guinea pigs from room air. Even when UVGI is confined to the upper room, good air mixing (ideally with low-velocity ceiling fans but easily accomplished by other types of forced-air ventilation) results in very high eACH in the lower, occupied space—estimated to be an additional 24 eACH in a South African study.⁴⁷ A similar study was conducted in Peru, where Escombe et al. found similar reduction in transmission.⁴⁹

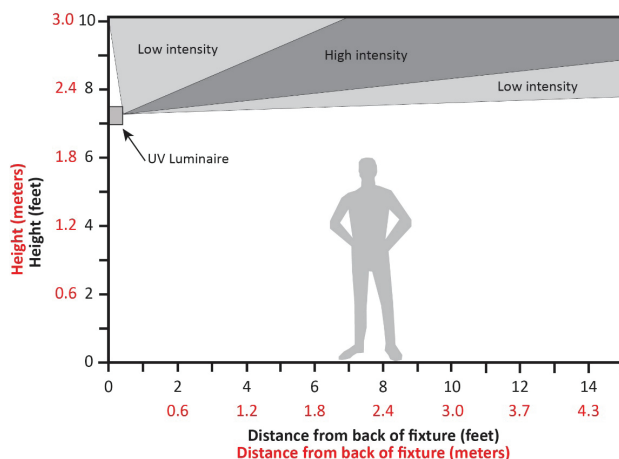


Figure 6-1. Upper-room GUV installation of an open luminaire.

When UVGI is used in ducts, although it ensures that recirculated air has a reduced level of viable pathogens, it unfortunately does relatively little to prevent person-to-person transmission in a room where both an infectious source and other susceptible persons share the same air. For effective interruption of transmission, air disinfection has to occur in the same room where transmission is occurring. Portable air cleaners can be placed in rooms

where there is a risk of transmission, but moving large volumes of air through any device is difficult, limited by the clean-air delivery rate of the portable air cleaner. Often when the clean air delivery rate is converted to eACH, the result is a disappointing 1 to 2 eACH, far too little to effectively prevent transmission. A large air cleaner in a small room may be effective, but for larger rooms, air cleaners are simply an impractical approach to high levels of air disinfection, compared to upper-room UVGI. Air cleaners may have value in a confined space where UVGI is desired.

There are very sophisticated programs to calculate the lamp source sizes and in-air dose requirements in terms of energy required for space and radiant fluence (joules per square meter, J/m²) across a cross-section of a UV-C beam. However, there is a much simpler, evidence-based dose that has been developed over many years for control of the bacterium that causes tuberculosis, typically specified as about 17 mW (0.017 W/m³) of 254-nm lamp-emission radiant power per cubic meter (m³) of irradiated space to disinfect the air.⁴⁷ Although this sounds too simplistic to be true, since air in any room is always moving and mixing, one can correctly assume that all air will be treated—the better the air mixing, the sooner this will happen. Studies at the Harvard School of Public Health^{47,48} and elsewhere show log units of reduction equivalent to 24 ACH to achieve 80% reduction of transmission. Of course, 100% reduction is not possible, because of the multiple modes of transmission.

Louvers are often used in fixtures to prevent people from viewing the UVGI source directly, to prevent eye damage (see **Section 5.3 Dangers to Humans**). However, with high-bay areas, high-ceiling warehouses, or high-ceiling marketplaces, a highly louvered UVGI fixture should not be required. Instead, much more efficient, properly installed *open-lamp* fixtures—wall mounted or suspended—are quite effective (see **Figure 6-2**). (Note: This was common in industrial sites many decades ago.) In a *louvered* fixture (see **Figure 6-3**), the UV radiation is directed mainly forward and slightly upward, as shown in **Figure 6-4**. Low-velocity ceiling fans and/or ventilation diffusers ensure good vertical air movement.

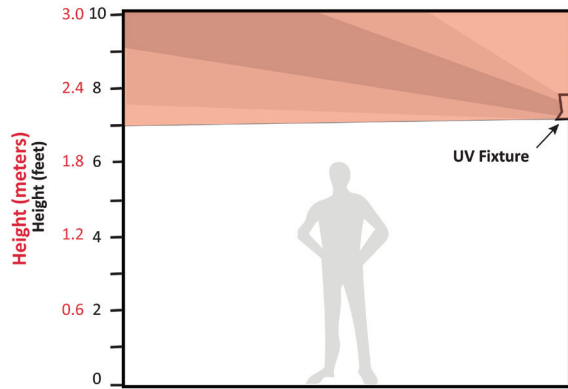


Figure 6-2. Open fixture with a forward UV-C energy distribution; for use in rooms more than 3 m (10 ft) in height. (Graphic © Illuminating Engineering Society)

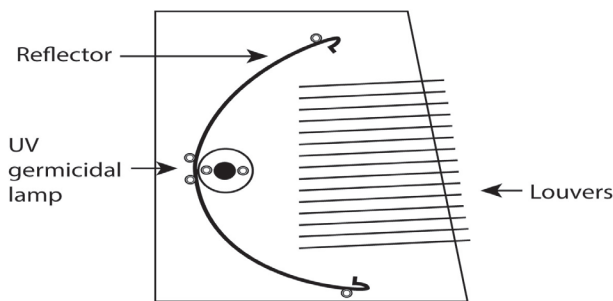
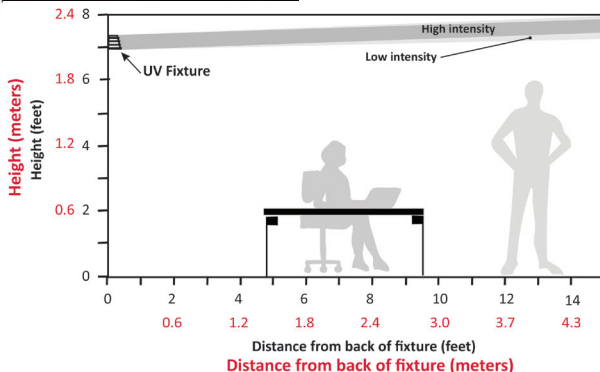


Figure 6-3. Example of a schematic cross-section of a louvered fixture. (© Illuminating Engineering Society)



Figure 6-4. Left and bottom: Output from a UV-C louvered fixture is emitted mainly forward and upward. Bottom: The fixture should be mounted at a minimum height of 2.1 m (7 ft). (Photograph courtesy of UV Resources; graphic © Illuminating Engineering Society)



It is important that upper-room UVGI air disinfection systems be planned, installed, tested post-installation for compliance (refer to **Annex A**), and maintained according to current standards or guidelines—for example, Chapter 62 of the *ASHRAE HVAC Applications Handbook*,⁵⁰ ACGIH's *2020 Threshold Limit Values (TLVs) and Biological Exposure Indices (BEIs)*,⁹ the End TB Transmission Initiative's document on GUV system maintenance,⁵¹ and two references from the United States Centers for Disease Control and Prevention.^{52,53} Only trained and knowledgeable professional should plan and install such systems. Room air cleaners, disinfecting air through HEPA filters, in-duct UVGI lamps, or other methods seem attractive, but their clean-air delivery rate when converted to room eACH is low—no more than 4 added eACH. (Generally, 6 to 12 eACH is required for adequate dilution of aerosolized pathogens.) UVGI in-duct air disinfection is a secondary approach to treating any recirculated air, albeit lacking the need for ensuring proper installation to prevent exposures above the TLV for all occupants in the lower part of the room.

Upper-room UVGI is still the safest, most effective application of UV-C to achieve significant room eACH. In special locations, where viral transmission is highly likely, whole-room UVGI (from suspended fixtures directing UV-C downward) could be applied, provided strict precautions can be followed (i.e., all unprotected occupants vacate the room). It is critical that any persons remaining in the space being disinfected from directly overhead and side exposure with UV-C lamps wear protective clothing and eye protection, or exposure to harmful UV radiation may occur. Whole-room UVGI has been safely applied in unoccupied rooms where entry is forbidden during the disinfection process. In the U.S., the Centers for Disease Control and Prevention (CDC) has provided guidelines for the use of UVGI lamps in upper rooms and air handling units (AHUs) as a supplemental control measure for air disinfection.^{52,53,54,55}

6.2 Room Surface and Object Disinfection

While germicidal UV radiation (UV-C in the appropriate range; see **Section 3.0 Types of UV Radiation**) is an excellent surface disinfectant, it does not penetrate surfaces and cannot disinfect soiled surfaces. The inability of the UV radiation to reach shadowed recesses of surfaces or to penetrate coverings like dust and other matter may negatively affect disinfection. For these reasons, it is critical that GUV lamps be regularly cleaned in order

to maximize their effectiveness, and UVGI is typically used as a supplemental control measure for disinfection. Maintenance personnel should follow the manufacturer's instructions for cleaning. A 2005 published study⁴⁶ concluded that UVGI could have some effect on the spread of infectious respiratory diseases, but there was inadequate evidence to support recommending its wide use. However, this result occurred in part because some of the general populations studied (e.g., elementary school students and military recruits in barracks) did not always practice other types of infection control in locations away from the GUV installations. For example, the authors of two school studies determined that close contact among the students on school buses and at play times negated some of the UVGI benefit and facilitated the spread of contagious disease.⁴⁶ The CDC recognizes that UVGI has several potential applications but also has limitations and possible safety issues.^{52,53,54} (Refer also to **Section 8 Safety Measures and Restrictions.**)

Hydrogen peroxide (H_2O_2)-vapor disinfection of surgical masks and N95 respirator masks is the most recommended method now in use. However, if this is not available, studies by several laboratories have shown surprisingly effective disinfection with UV radiation despite the fact that UV photons will not have a straight-line pass through all of the porous filter structure. Therefore, forward-scattered photons have to penetrate the mask, and substantial doses are required.⁵⁶ This should only be attempted within a light-tight enclosure. In the U.S., NIOSH, the FDA, and the NIH have issued temporary guidance on this important subject.^{53,57,58,59}

The amount of UV-C needed to disinfect surfaces depends on the targeted pathogen as well as the type of surface and its cleanliness; recommended exposures vary from 200 to 1,000 joules per square meter (J/m^2) (20 to 100 mJ/cm^2).⁶⁰

Hand-held, compact UVGI products are available. Most of these emit relatively little UV-C, making it difficult to attain reliable disinfection. Additional information is provided in **Section 4.5 Handheld Devices.**

In summary, while UV-C could be a secondary infection control measure for disinfecting potential germ-carrying deposits on accessible (not-shadowed) room and object surfaces, its great value would be in disinfecting air in

areas where this may be a concern (e.g., intensive care wards, hospital intake facilities [or tents]). **Section 6.1 Air Disinfection** provides additional information.

7.0 Maintaining Systems and Verifying Performance

It is important to keep UVGI systems in good working order, including checking them periodically to be sure they are still performing as required. There are a number of dedicated meters available for measuring UV irradiance; however, a wide range of scale is normally required, e.g., a range from 0.1 to 100 microwatts per square centimeter ($\mu W/cm^2$) for safety.²¹ Safety readings require the lower range, and efficacy requires a range up to at least 10 milliwatts per square centimeter (mW/cm^2). It is important that the meter be suitable and calibrated for the measured wavelengths. Meters optimized for 254 nm will likely measure emissions from LED or KrCl* lamps with large errors. A common practice is to have two calibrated meters: one in reserve and for reference. The two instruments should be periodically compared. The user should retain the manufacturer's instructions, including a description of the meter, its safe use, and maintenance and calibration of it. Some healthcare facilities contract with a full, outside maintenance contractor that uses calibrated meters and correctly and safely replaces burned-out lamps. Some users retain a simple, less precise meter for staff to use, but the installer uses a professional meter.⁵¹

The American Society of Heating, Refrigeration, and Air-Conditioning Engineers (ASHRAE) has published two method-of-test standards to evaluate UV equipment for air use in air ducts:

- *ASHRAE Standard 185.2-2014, Method of Testing Ultraviolet Lamps for Use in HVAC&R Units or Air Ducts to Inactivate Microorganisms on Irradiated Surfaces*¹⁷
- *ASHRAE Standard 185.1-2015, Method of Testing UV-C Lights for Use in Air-Handling Units or Air Ducts to Inactivate Airborne Microorganisms*¹⁹

The measurement of UV radiation is best performed by an experienced commissioning or maintenance company, trained in exposure limits and with meters calibrated

for this task (refer to **Annex A – Commissioning Upper-Room UVGI Disinfection Systems**).

8.0 Safety Measures and Restrictions

8.1 Safety Rules for Air Disinfection With UVGI

To ensure the safe use of UVGI equipment for *air* disinfection, these guidelines should be followed. These are described in the subsections that follow.

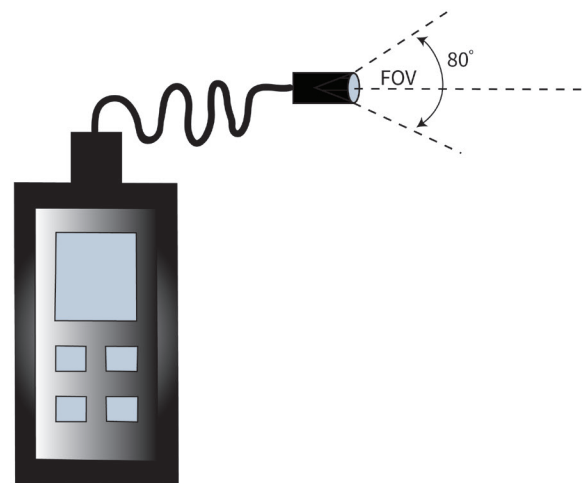
8.1.1 All GUV Sources. Workers should place warning signs near upper-room UVGI sources and on AHU access panels where internal UVGI sources are installed. Activation switches should be clearly labeled and protected with switch guards to prevent accidental activation by unauthorized personnel. If exposures cannot be avoided, workers should wear plastic or glass face shields to protect the eyes and face, nitrile gloves or work gloves to protect the hands, and full-coverage clothing with tightly woven fabrics to protect all other exposed skin.

It is important that occupational safety measurements be made after the installation of GUV systems for air disinfection in a room that will ultimately be occupied; they should be made using a radiometer with a detector calibrated for the wavelength selected. If the TLV at eye level will be exceeded over an 8-hour period, adjustments need to be made before the system is turned on for continuous occupancy.

8.1.2 Upper-Room GUV Sources. Proper installation is critical to ensure the safe use of these UVGI sources. It is important that ceiling reflectance at the UV-C wavelength be studied before installing in a new location, since down-welling UV-C could be increased.²² Regular work activity should not resume in rooms with upper-room UVGI equipment unless qualified measurements have confirmed that the potential radiant exposures in the lower room are within the 8-hour exposure limits specified by the ACGIH.⁹ It is critical to note that persons who may be exposed over a full day (e.g., hospital patients in isolation wards with upper-room UVGI installations, who may be exposed

to low levels of these emissions over a 24-hour period) should not receive exposures that exceed the ACGIH 8-hour TWA exposure limit of $0.1 \mu\text{W}/\text{cm}^2$ effective (i.e., $0.2 \mu\text{W}/\text{cm}^2$ at 254 nm). Except in very large rooms, the emitting UVGI sources should not be directly visible through louvers to occupants in the lower room.

Commissioning an upper-room installation requires expertise and special measurements. The UV-C measuring instrument should be calibrated for use with the type of UV-C source (e.g., Hg, KrCl*, LED) and emitted wavelengths, and fitted with an 80-degree (full-angle) FOV cone, as shown in **Figure 8-1**. **Figure 8-2** shows the proper positioning and orientation of the FOV cone for determining the direct downward reflections from the ceiling to ensure safe skin exposure of a standing person; the FOV cone is aimed toward the fixture(s) at various points at least 1.8 m (about 6 ft) above the finished floor (AFF), the position of the eyes of a tall standing person. (Refer to **Annex A** for additional information.) Although the 80-degree cone applies to evaluating both potential skin and eye exposure, for assessing the eyes' exposure, the cone would more likely be directed horizontally or even slightly downward, to simulate the direction of a person viewing a task expected in that occupied space. Time-weighted averaging of each type of exposure applies.



Safety meter with 80° cone FOV

Figure 8-1. Instrument for measuring UV-C radiation should have an 80-degree field of view (FOV).

(Graphic © Illuminating Engineering Society)

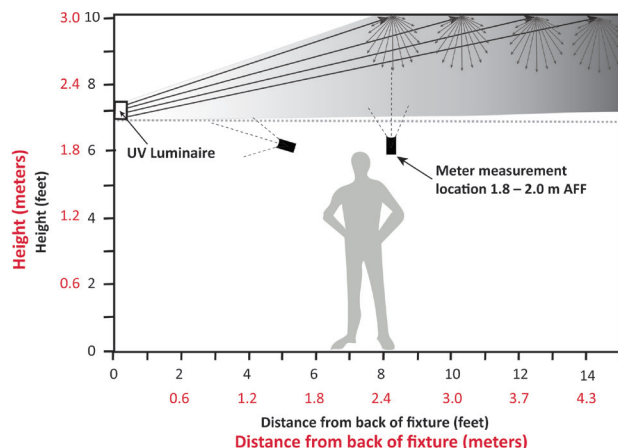


Figure 8-2. Showing the positioning and aiming of meters used to measure UV radiation in an upper-room installation. (Graphic © Illuminating Engineering Society)

Figure 8-3 shows how various exposures are averaged over time to achieve what is referred to as a time-weighted average exposure. **Figure 8-4** shows an example of how, with appropriate time-

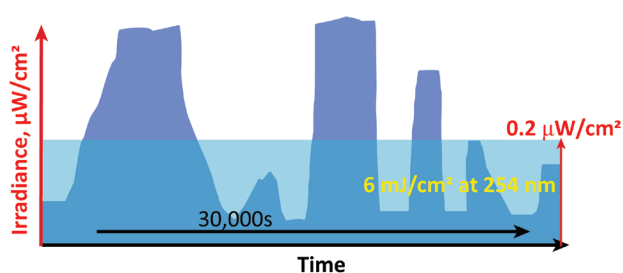


Figure 8-3. An example of time-weighted averaging. (Graphic © Illuminating Engineering Society)

weighted averaging, the TWA irradiance could be different for persons seated (as in a conference room or office), compared to those who might be standing; in this case, a conservative worst-case duration for reaching the TWA in a standing position was determined as 1 hour, compared to 8 hours for seated persons. The proper application of the TLV requires an estimation of the actual TWA irradiance and not just the maximum value.

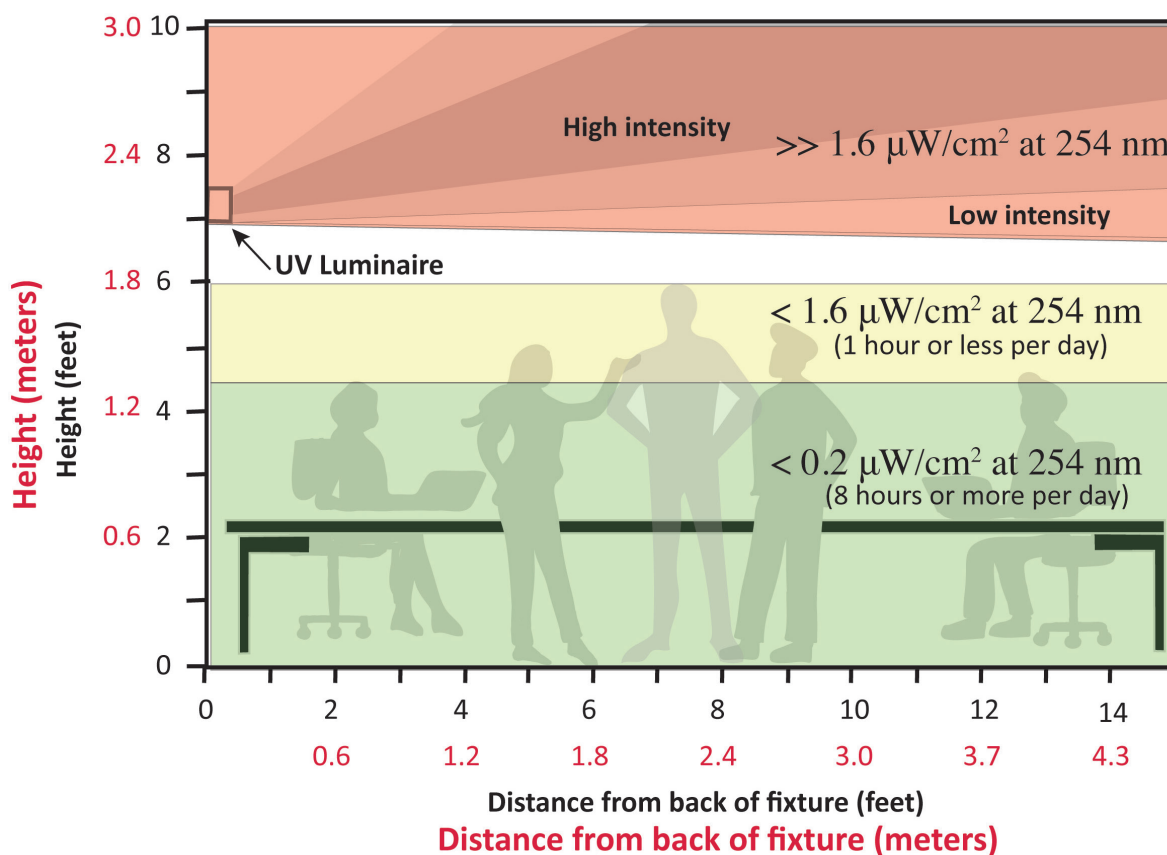


Figure 8-4. Example of how a time-and-motion study led to different TWA values for eye exposures at seated and standing positions. (Graphic © Illuminating Engineering Society)

Figure 8-5 provides examples of two installations that were not commissioned; in each case, actual accidental overexposures led to eye irritation (photokeratitis), which although transient (lasting a day) created discomfort, resulting in very annoyed individuals. The left graphic illustrates how a person going down the stairs was in direct line of sight of the lamp. The right graphic illustrates how the installer misunderstood that the fixture was to be aimed *upward* according to the manufacturer's instructions—rather than downward—and an incidence of photokeratitis eventually occurred.

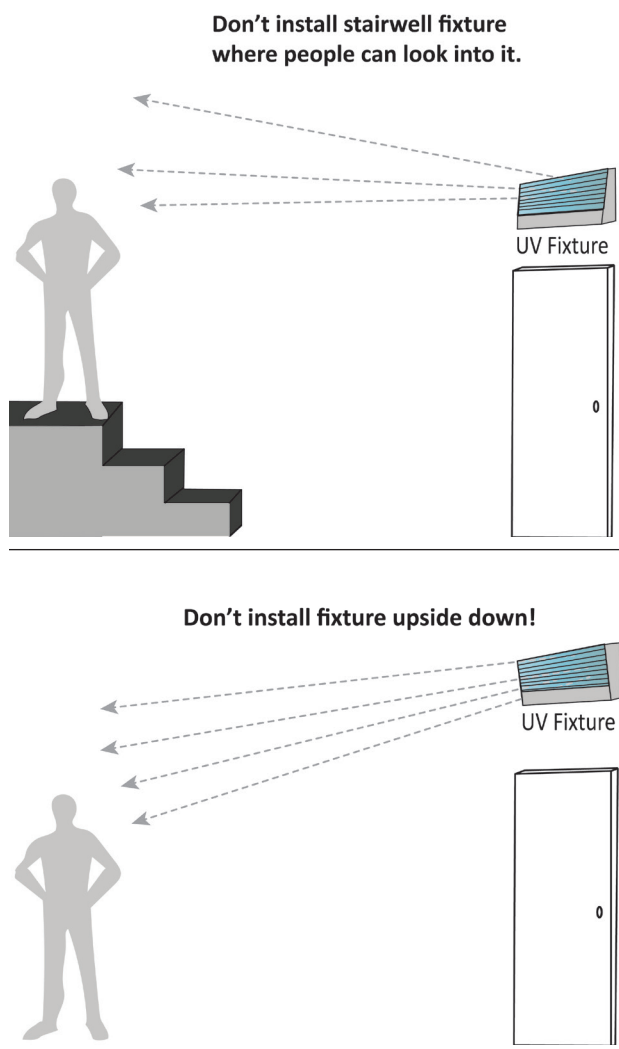


Figure 8-5. Examples of incorrect installations, showing why commissioning of UV-C equipment installations is critical. Top: A stairwell installation where people can see into the fixture; bottom: incorrect fixture aiming (should be aimed *upward*, not downward). (Graphics © Illuminating Engineering Society)

8.1.3 AHUs with Internal GUV Sources. Access panels for AHUs with internal UVGI sources should be interlocked with automatic shutoff switches to prevent accidental exposure to UV radiation. An inspection window that blocks germicidal UV energy (e.g., plastic or glass) should be installed to allow workers to see whether the UVGI source inside the AHU is operating.

8.2 Safety Rules for Surface Disinfection With GUV

To ensure the safe use of GUV equipment for *surface* disinfection, these guidelines should be followed:

- **All GUV sources.** Workers should place temporary warning signs at access points to the area being disinfected. They should either vacate the area during disinfection or place barriers opaque to UV-C between the GUV source and room occupants. If these areas are required to be occupied during disinfection and exposures cannot be avoided (e.g., if using a handheld disinfecting GUV “wand”), then personal protective equipment (PPE) should be used when there is a potential for exposure above the TLV. Workers should wear plastic or glass face shields to protect the eyes and face, nitrile gloves or work gloves to protect the hands, and clothing with tightly woven fabrics that covers other exposed skin.

As direct GUV systems for surface disinfection systems are added in various spaces, occupancy sensors should be used to ensure that the systems are deactivated when the room is occupied, unless programmed for specific medical applications.⁶¹

- **Low-pressure Hg lamps, GUV LEDs, and most other UV-C sources.** Workers should wear plastic or glass face shields to protect the eyes and face, nitrile gloves or work gloves to protect the hands,⁶² and full-coverage clothing with tightly woven fabrics to protect all other exposed skin.
- **Pulsed xenon arc lamps.** Only trained personnel should operate these devices and should follow the manufacturer's procedures. Workers should wear welding or cutting goggles to protect the eyes, nitrile gloves or work gloves to protect the hands, and full-coverage clothing with tightly woven fabrics to protect all other exposed skin.

8.3 Safe Viewing During Room Disinfection

If a room has glass windows that permit vision into a room and a view of its UV fixtures, the glass does not need to be shuttered or covered for safety if the fixtures have 254-nm UV-C lamps. Glass windows block potentially hazardous UV-B and UV-C transmission. Glass windows should be covered if pulsed xenon lamps are in use.

8.4 Using UVGI in Hospital Tents

Press reports show installation of UV-C lamps in hospital tent ceilings (whole-room UVGI) where all persons were fully protected by infection-control head and face coverings, gloves, and coveralls. This should have been safe if all complied with complete protective covering. Upper-room disinfection should not be ruled out if hospital tents are quite high and sufficient airspace above 2.1 m (7 ft) is available (see **Section 6.1 Air Disinfection**), but the special upper-room UVGI fixtures would have to be very stable so as not to tip and thereby direct UVGI downward into occupied spaces. It is also important to note that proper air circulation is critical to the effectiveness of upper-room disinfection.

8.5 Use of GUV Equipment in the Home

As described in **Section 4.5**, handheld, compact UVGI products (see **Figure 4-4**) are sold but are considered a serious safety concern in a general household environment, where children, pets, or careless adults can easily be overexposed. These products are typically less than 10 watts, with open or exposed mercury lamps or UV-C LEDs. They may come with a safety timer; a person places the open lamp on a table or in a convenient location, sets the timer for several minutes to an hour, and is given a 10-second delay to quickly exit the room and close the door. Because the vast majority of consumers are inexperienced regarding the potential hazards of UVGI, people who purchase these devices should use them with extreme caution.⁶³

It should be noted, however, that the employment of UVGI equipment within a home HVAC system is a standard, safe application for reducing mold, especially in humid climatic conditions (refer to **Section 4.2 In-Duct Systems – Surface and Air**).

Finally, customers interested in purchasing “disinfection” tools that emit UV-C radiation will find that the internet is full of such products. The validity of the product claims is often questionable, so caution is advised for such purchases.

9.0 Overview: Systems Comparisons

9.1 Options and Basis for Application Type

Ultraviolet germicidal irradiation (UVGI) applications can be used as a valuable component of a multi-tiered approach to reduce the spread of infectious diseases. This section provides recommendations for the existing types of UVGI technologies for use in various infection control applications, first evaluated by the type of application (**Section 9.2**), by the type of pathogen (**Section 9.3**), and finally by the means of pathogen transmission (**Section 9.4**). There are clear advantages and disadvantages related to safety and efficacy for each type of technology. **Section 9.5** provides a tabular overview.

9.2 Evaluated by Type of Application

Upper-room (indirect) UVGI: Although ultraviolet LEDs are entering the market quickly, most upper-room GUV fixtures currently employ low-pressure Hg lamps in wall-mounted or suspended fixtures—louvered for ceilings below 2.4 m (8 ft), un-louvered for higher ceilings. UV LED technology is being used in the same way and offers greater fixture design flexibility, no Hg waste, and possibly longer lamp life.

Advantages: Air disinfection occurs in the room where the transmission risk exists. Combined with low velocity ceiling fans and/or ventilation diffusers to ensure good air mixing, upper-room UVGI provides safe and effective whole-room air disinfection that has been shown to be as much as 80% effective in two controlled clinical trials by adding the air disinfection equivalent of 24 ACH in addition to an existing 6 ACH—the equivalent of 1 air change every 2 minutes! This is well beyond most ventilation systems or room air cleaner air disinfection rates.

Disadvantages: Sufficient ceiling height is required for safe and effective application. Fear of occupant exposure to UV radiation, even though it has been shown to be entirely safe when properly applied, is a major barrier to acceptance. There is no evidence of significant surface decontamination from upper-room UVGI, although some surface effect from continuous low level UV exposure is likely.

UV in-room air cleaners (portable or installed):

Where ceiling height is too low for upper-room UVGI, or for very small rooms, a portable air cleaner may be used with HEPA filtration, GUV emission, or both.

Advantages: Air disinfection occurs in the room where transmission risk exists. Effectiveness depends on the rate of air disinfection. Portable air cleaners are rated by clean air delivery rate (CADR), equivalent to the fresh (not recirculated) ventilation rate. Properly sized for the room, a well-designed room air cleaner can deliver six or more eACH. They require no installation.

Disadvantages: Manufacturers of room air cleaners often tout as evidence of efficacy the single-pass removal or inactivation of test microorganisms. Usually, UV emission, HEPA filters, or both remove 99% to 99.9% of pathogens. However, this is not the critical parameter; it is the CADR relative to the size of the room. Often, the CADR results in less than 1 or at most 2 eACH, compared to the 20 or more eACH possible with upper-room UVGI.

Whole-room direct-UVGI air and surface disinfection in occupied rooms (222 nm): Because it appears to pose a very low risk for human skin and eyes if properly used, but is highly germicidal, 222-nm UVGI may become the optimal way to disinfect room air and surfaces, especially for pathogens like SARS-CoV-2 (which causes COVID-19) that transmit via both air and surfaces. Although there are not yet clinical trials comparable to upper-room 254-nm UVGI, direct 222-nm UVGI may ultimately be shown to be as effective as or more effective than upper-room UVGI, especially for COVID-19.

Advantages: Air disinfection occurs in the room where the transmission risk exists. With proper commissioning (e.g., occupant TWA UV-C exposure assessment and ozone considerations; refer to **Annex A** for additional

information), occupant safety can be ensured. Surface decontamination, especially of clean surfaces, is possible.

Disadvantages: Cost and availability are currently major barriers. Safety studies are still being done, although all results thus far suggest safety when used within guidelines. Short lamp life for most lamps adds to the expense. Like other UVGI surface disinfection, ceiling-mounted UVGI is unable to disinfect shadowed areas. Some lamps are operated intermittently to extend lamp life, which may allow some pathogens to spread between room occupants during the lamps' inactivation periods.

Whole-room direct UVGI surface disinfection in unoccupied rooms (Hg 254 nm; pulsed xenon lamps, stationary or portable; or moving robots):

This technology has been long used in Russia, and more recently in the U.S., for decontaminating rooms to prevent hospital-acquired infections such as methicillin-resistant *Staphylococcus aureus* (MRSA) and *Clostridium difficile* between patients, thus complementing surface cleaning.

Advantages: It has been shown to be effective when used in addition to cleaning and chemical decontamination.

Disadvantages: it is expensive. UVGI is inherently imperfect for surface decontamination because photons travel in straight lines and will miss shadowed areas. Moving robots help eliminate shadowing effects, but not entirely. This is not useful for airborne infection control because it cannot be used in occupied rooms where transmission risk exists.

Whole-room direct-UV air and surface disinfection in occupied rooms, with occupants wearing UV-protective clothing and eye protection:

This has been used in orthopedic-surgery spaces to prevent infection of appliances such as artificial joints. Although this approach has proven highly effective in preventing infections that require appliance removal, experience has shown that it inevitably is discontinued due to occupant UV injury resulting from breaking protocol—by removing goggles, for example. UVGI of 222 nm is being used in Japan for orthopedic surgery without the

risk of eye or skin damage.⁶⁴ A similar approach using UVGI at 254 nm has been suggested for preventing COVID-19 transmission, but thus far it has not been implemented and reported in the literature.

Advantages: Direct air and surface disinfection at 254 nm has proven highly effective in prevention of orthopedic wound contamination. Current research suggests that 222-nm UVGI may be safer for preventing orthopedic wound contamination and for implementing infection prevention and control than is 254-nm UVGI.

Disadvantages: Safety concerns require strict protocol adherence. Direct GUV exposure is not widely practiced in operating rooms because the strict protocol adherence can be difficult to sustain, and strict adherence would be less likely in other hospital settings. Although superficial eye and skin UV injury is much less serious than contracting COVID-19, there are good alternatives to using 254-nm UVGI in occupied rooms directly with protective clothing and eye protection: indirect upper-room UVGI and 222-nm direct UVGI without the need for protective clothing.

UVGI in HVAC duct work to prevent recirculation of airborne pathogens: UVGI is used in HVAC systems in at least two ways: fungal decontamination of cooling coils and drip pans, and air irradiation in return ducts to prevent recirculation of pathogens. Both are long established applications of proven efficacy and safety.

Advantages: Use in ducts without human exposure allows for use of high doses that are highly effective for various pathogens, based on long-established application guidelines with respect to duct area and air flow rate. UVGI is also highly effective on cooling coils and drip pans.

Disadvantages: For some infections with known spread through the HVAC system, return-duct UV air disinfection may be useful. Since lamps are normally not routinely observable, timely maintenance can be difficult.

9.3 Evaluated by Type of Pathogen

The application of germicidal UV technology is determined by the characteristics of specific pathogens, but can also be designed to cover a wide range of

the most likely types of pathogens. Several types of pathogens are discussed in this section.

Water borne microbes: Germicidal UV is widely used for water disinfection. It is considered a highly effective use but is not discussed further here.

Surface borne microbes: For common hospital-acquired (nosocomial) pathogens such as staphylococcus and *C. difficile*, whole-room surface disinfection in unoccupied rooms between patients is an established practice.

Airborne microbes: For respiratory pathogens such as influenza, tuberculosis (TB), and SARS-CoV-2, transmission patterns dictate GUV applications. TB is an exclusively airborne pathogen that spreads within rooms, and occasionally between rooms through ventilation ducts, but not from surfaces. The dose (quanta of bacilli) required for TB infection is believed to be relatively small, permitting common transmission within rooms, but less common transmission after dilution in room air or ventilation ducts. This mode of transmission dictates air disinfection within the room where transmission is likely, using indirect 254-nm Hg, 265- to 280-nm LED, or possibly direct/indirect 222-nm KrCl* lamps. The use of in-duct UVGI is standard practice for reduction of pathogens in ventilation systems, since in many healthcare settings air may be recirculated many times. In-duct UVGI is a common strategy as a bioterror countermeasure.⁶⁵ In-duct use of UVGI has shown a benefit for office workers' well-being and improved indoor air quality.^{66,67}

GUV in-room air cleaners are used occasionally when ventilation improvements are not practical because of room height.

Influenza and SARS-CoV-2 spread through a combination of large and small respiratory droplets and through contaminated surfaces. For COVID-19, current epidemiological evidence favors transmission within rooms rather than between rooms,⁶⁸ suggesting that a relatively large dose of virus is required for infection. Social distancing from the source not only prevents direct contact with large respiratory droplets and surfaces contaminated by large respiratory droplets, but also reduces infection by small airborne droplets

through dilution. Capture in the room exhaust results in further dilution, as return air mixes with other return air before it is filtered and returns to other occupied rooms. Probably for this reason, there is currently little evidence of COVID-19 transmission through ventilation ducts to warrant a focus on enhanced filtration. Indeed, more-efficient filters offering greater flow resistance may reduce net ventilation rate, thus increasing rather than decreasing risk.

In general, there may be several pathogens circulating in public spaces, homes, and healthcare institutions; for example, during the COVID-19 pandemic there are parts of the world where pathogens causing tuberculosis, influenza, SARS-CoV-2, and measles all could be circulating in the same space. In these cases, selecting the type of GUV system or approach will depend on the most resistant strains of microbes needing inactivation. This is why TB is a good target pathogen for consideration of air methods. For surface decontamination, *C. difficile* and adenovirus could be good targets.

9.4 Evaluated by Means of Pathogen Transmission

Airborne transmission:

- Primary applications for reducing airborne transmission would include upper-room GUV (254 nm) systems, which are well known. (See **Tables 9-1a** and **9-1b**.) This environmental control can supplement ventilation (natural, mechanical, or hybrid systems).
- Secondary application for airborne transmission in recirculated air would be to use GUV (254 nm) systems in the air handling unit and ductwork of an existing HVAC system. (See **Tables 9-2a** and **9-2b**.)
- A third method would be to use a high-quality room air cleaner with UVGI (222 or 254 nm); however, this should be considered only if room geometry will be either of the two recommendations above. (See **Tables 9-3a** and **9-3b**.)
- A fourth method, for consideration in a pandemic only, would be to install a whole-room UV-C system such as an open UV-C fixture (254 nm) at a height greater than 3.7 meters (about 12 feet). This would irradiate both air and surface but would require personal protection equipment (PPE) for occupants.

Because of the known risk of GUV exposure, open-fixture GUV systems should only be utilized in areas where other upper-room GUV applications, described above, cannot be deployed. (See **Tables 9-4a** and **9-4b**.)

Surface-borne transmission:

- The primary approach for surface disinfection would be the use of autonomous mobile UV-C devices (“robots”) or permanently mounted UV-C devices in unoccupied spaces in conjunction with prior manual cleaning of patient rooms. (See **Tables 9-5a** and **9-5b**.)
- A secondary approach would be the use of wall mounted fixtures that have an upward component for air as well as a downward component, with the UV-C sources controlled by an occupancy sensor.
- A primary approach for in-duct cleaning is to use UV-C sources on cooling coils to reduce the growth of mold. (See **Tables 9-6a** and **9-6b**.)

General information regarding the tables in Section 9.5:

The first column, “Source,” indicates the type of UV-C source.

The next three columns refer to the relative level of application of this source in a particular system. “Airborne” refers to the source’s application for disinfecting airborne bacteria and viruses; “Surface-Borne” refers to the source’s application for disinfecting surfaces contaminated with bacteria and viruses; and “Air and Surface” refers to the source’s application for disinfecting both the air and surfaces contaminated with bacteria and viruses.

The “Evidence Based” column refers to the qualitative level of published studies showing this source’s effectiveness, while the “Experimental” column refers to the qualitative level of current research on this particular application.

“Use During Pandemic” and “Use Any Time” refer to the subjective recommendation of whether this source and system combination should be used in pandemic situations and under any conditions, respectively.

9.5 Summary Tables

The tables in this section are described and explained in **Section 9.4**.

Table 9-1a. Summary of Upper-Room GUV Air Disinfection Systems, Louvered and Open

Source	Airborne	Surface-Borne	Air and Surface	Evidence Based	Experimental	Use During Pandemic	Use Any Time
Hg lamp systems	*****	N.A.	N.A.	*****	N.A.	✓	✓
UV-C LED	** [1* → 2*]	N.A.	N.A.	*	****	✓	✓
Excimer lamp	*** [4* → 3*]	*	N.A.	**	****	✓	✓

Table notes:

- Asterisks (*) rate the primary benefit; a greater number indicates a stronger recommendation.
- N.A.: Not applicable or no evidence.

Table 9-1b. Advantages and Disadvantages

Advantages	Disadvantages
Mercury (Hg) systems	
• Suitable for all temperate or controlled climates and ventilation conditions • Achieves high levels of equivalent air changes per hour (eACH) • A built-in solution for room air disinfection • Disinfects large volumes of room air when combined with forms of mechanical ventilation, natural ventilation, or hybrid types to mix the air; may have a secondary benefit for surface decontamination • 70+ years of evidence-based experience; can be safely deployed and effectively used with proper installation, commissioning, operation, maintenance, and facility staff training (see Annex A) • Used extensively to mitigate the airborne spread of tuberculosis • Comes in a variety of open and louvered fixtures for different room geometries • Recommendations and guidelines available	• Can be expensive to install and maintain • Potential occupational exposure from poor installation, improper maintenance, or human error • Not widely available on a global basis • Requires expertise to design, commission, install, operate, and maintain (see Annex A) • Limited global maintenance companies • Requires a minimum ceiling height of 2.4 m (8 ft) • Some difficulty in disposing of Hg lamps
LED UV-C systems	
• Expanded options with UV LEDs in various wavelengths • Good optical design • Basic guidelines for Hg systems may be used	• Expanded options in various wavelengths will create a need for greater awareness of eye and skin safety concerns in the 265-nm to 280-nm range • Heat needs to be dissipated • Cost • Output and efficiency
Excimer systems	
• Expanded options for available wavelengths • Basic guidelines for Hg systems may be used • Low penetration of skin may increase human safety for exposure	• Commercially available sources are limited • Lamps may need to be filtered, especially if some human exposure is expected • Short lamp life: 3,000 to 4,000 hours, with some untested claims of 30,000 hours • Eye safety may be a concern • Heat needs to be dissipated • Cost • Output and efficiency

Table 9-2a. Summary of GUV HVAC Air Disinfection Systems

Source	Airborne	Surface-Borne	Air and Surface	Evidence Based	Experimental	Use During Pandemic	Use Any Time
In-duct UV-C (Hg enclosed in-duct or in air-handling units)	*****	*****	*****	****	N.A.	✓	✓

Table notes:

- Asterisks (*) rate the primary benefit; a greater number indicates a stronger recommendation.
- N.A.: Not applicable or no evidence.

Table 9-2b. Advantages and Disadvantages

Advantages	Disadvantages
• Suitable for all temperature or controlled climates • Enclosed system with required safety interlocks • Treats supply air or recirculated air • Treats cooling coils to keep clean of mold and biofilm	• Can be expensive to install and maintain in resource-limited settings • Requires UV-C-safe filter media, if filters are exposed

Table 9-3a. Summary of Other UV-C-Plus-Filter (Enclosed) Room Air Cleaner Systems

Source	Airborne	Surface-Borne	Air and Surface	Evidence Based	Experimental	Use During Pandemic	Use Any Time
Hg lamp; UV-C LED	***	N.A.	N.A.	*	*	?	?

Table notes:

- Asterisks (*) rate the primary benefit; a greater number indicates a stronger recommendation.
- N.A.: Not applicable or no evidence.
- Question mark (?): Effectiveness is limited, based on clean-air delivery rate and room volume.

Table 9-3b. Advantages and Disadvantages

Advantages	Disadvantages
• Suitable for all temperate or controlled climates • Enclosed unit • Portable	• Quality can vary widely • Can require costly filter media (if HEPA system is used) • Clean-air delivery rate is generally very low in residential air cleaners

Table 9-4a. Summary of Whole-Room UVGI Disinfection Systems (Requires PPE for Occupants, or an Unoccupied Room with Occupancy Sensor)

Source	Airborne	Surface-Borne	Air and Surface	Evidence Based	Experimental	Use During Pandemic	Use Any Time
Hg lamp suspended from the upper room	***	****	**** [3* → 4*]	****	*	✓	✓ (primarily medical)
UV LED systems	***	***	***	N.A.	****	✓	✓ (primarily medical)
Excimer lamp	*****	*	N.A.	**	****	✓	✓ (primarily medical)

Table notes:

- Asterisks (*) rate the primary benefit; a greater number indicates a stronger recommendation.
- N.A.: Not applicable or no evidence.

Table 9-4b. Advantages and Disadvantages

Advantages	Disadvantages
• Suitable for pandemic conditions for air and surface • Bare lamps can be used; only minimal electrical connection needed • Relatively inexpensive for wide-area surface and air disinfection	• Requires occupants to wear personal protective equipment for eyes and skin in occupied areas • Possible material degradation • Would require occupancy sensors if the use of personal protective equipment cannot be ensured

Table 9-5a. Summary of Mobile UVGI Whole-Room Bare-Lamp Surface Disinfection Systems (Room Unoccupied)

Source	Airborne	Surface-Borne	Air and Surface	Evidence Based	Experimental	Use During Pandemic	Use Any Time
Hg lamp; pulsed xenon lamp; industrial excimer lamp	N.A.	*****	N.A.	**** [3* → 4*]	N.A.	✓	✓

Table notes:

- Asterisks (*) rate the primary benefit; a greater number indicates a stronger recommendation.
- N.A.: Not applicable or no evidence.

Table 9-5b. Advantages and Disadvantages

Advantages	Disadvantages
• Suitable for all temperate or controlled climates and ventilation conditions • Achieves high levels of surface decontamination and be used to enhance terminal cleaning of patient rooms upon discharge • Can be deployed throughout a facility for targeted disinfection	• Very expensive • Requires staff training • No method to benchmark output • Possible material degradation

Table 9-6a. Summary of GUV HVAC Surface Disinfection Systems

Source	Airborne	Surface-Borne	Air and Surface	Evidence Based	Experimental	Use During Pandemic	Use Any Time
In AHU by coil or filter (Hg lamp); unfiltered excimer lamp	**	****	***	****	N.A.	✓	✓
UV-C LED, 220 – 270 nm	N.A.	****	N.A.	*	****	✓	✓

Table notes:

- Asterisks (*) rate the primary benefit; a greater number indicates a stronger recommendation.
- N.A.: Not applicable or no evidence.

Table 9-6b. Advantages and Disadvantages

Advantages	Disadvantages
Mercury (Hg) Systems	
• Less UV-C energy required than for air disinfection systems • Suitable for all temperate or controlled climates • Enclosed system with required safety interlocks • Treats supply air or recirculated air • Treats cooling coils to keep them clean of mold and biofilm • Coil fins made of aluminum enhance UV-C distribution into the coil	• Can be expensive to install and maintain in resource-limited settings • Requires UV-C-safe filter media • Possible material degradation
LED UV-C Systems	
• Suitable to all climates • Small sources, lending to compact, targeted decontamination of high-touch areas • Can be built-in and controlled by occupancy sensors • Low power requirements • Output can be varied by using with a dimmer • Multiple wavelengths to choose from • Instant-on	• Expensive • Output for large-scale disinfection is limited • Gaps in data on effectiveness at various wavelengths • Significant risk of injury if used in consumer products • Possible material degradation • Thermal management required

10.0 Additional Information

Additional information on photobiological safety is available from IES.⁸ A recorded webinar, "Introduction to Ultraviolet and Visible Radiation Disinfection," is also available.⁶⁹

In addition to these resources from IES and those in the **References** section, there are a number of scientific research papers relating to GUV, for those interested in studying the subject in much greater detail.⁷⁰⁻⁸⁹

Annex A – Commissioning Upper-Room UV Disinfection Systems

This Annex is not part of ANSI/IES RP-44-21, Recommended Practice: Ultraviolet Germicidal Irradiation (UVGI). It is provided for informational purposes only.

ANSI/ASHRAE/IES Standard 202⁹⁰ describes the commissioning process, the roles of the principal agents and stakeholders, and a framework for developing design documents, specifications, procedures, documentation, and reports. Specifically, the Standard includes:

- Overview of commissioning process activities
- Description of each process step's minimum activities
- Minimum documentation requirements
- Acceptance requirements.

The owner's project requirements (OPR) are developed by the owner(s) and the commissioning agent during the pre-design phase. At a minimum, the OPR should include:

- Facility use and operating schedule
- Energy and efficiency goals
- Indoor space requirements such as temperature and ventilation
- List of all systems and equipment to be commissioned
- Project schedule

- Testing and sampling procedures
- Training requirements
- Submittal requirements
- Commissioning report format

The final acceptance testing (one of many steps in the commissioning process) of a new upper-room installation is to make sure the lamps are operational and to ensure that no one in the occupied zone is exposed to more than the applicable TLV, a multistep checklist may be followed:

Step 1: Check the Disinfection Zone. Begin at the center of the room. On the UV-C meter, remove the 80-degree cone field-of-view (FOV) clip-on to expose the full cosine detector. With the detector on a long pole, perform a simple check-test of the upper-room germicidal zone to be certain that the fixtures are properly working. In this step, the detector is oriented toward each fixture to measure the total irradiance at the applicable wavelength (e.g., 222 nm 254 nm, or applicable wavelength, depending on the lamp). It is important to note whether the meter is directly reading in radiometric units or is $S(\lambda)$ -weighted in order to compare with the applicable TLV (refer to **Section 2.1**, definition for Threshold Limit Value). The irradiance reflected from the ceiling onto a horizontal plane in the room should also be measured, again with the FOV cone removed from the cosine detector head. Be cautious about any attempt to estimate the "average upper-room irradiance" recommended by some experts, since that really should have been executed in the design phase.

A more thorough approach would be to check for the maximal reading by walking away from a single fixture (with any adjacent fixtures switched off), along its centerline of emission, raising and lowering the stick with the meter on it. Repeat with all fixtures operating, taking a measurement toward, as well as away from, each fixture. Then check for reflections off various surfaces. Finally, if adjacent rooms are fitted with fixtures, check for external contributions through passageways of interest.

Step 2: Standing Occupant Safety. Now place the 80-degree safety cone on the detector head to perform the safety measurements.

To assess eye exposure: Perform the test at the highest reasonably accessible position of a human eye for a standing occupant. Outside of a basketball court or similar facility where the eye level of a standing person could be well above the 99th percentile, the irradiance is usually measured in a vertical plane at about 1.8 m (6 ft) above the floor. Aim the axis of the cone toward the wall to measure any radiant energy emitted nearly parallel to the floor. The detector axis should lie in a horizontal plane (to receive irradiation on a vertical plane representing the eye). It should be aimed in each cardinal direction sequentially around 360°, taking into consideration the likely viewing duration of a person in that space and setting to be sure the irradiance is less than the maximum permitted. Some experts suggest using a one hour as a maximum reasonably foreseeable duration for a standing person, but this could vary with the setting. If uncertain, it may be necessary to observe the activities of the occupants at different times on different days in order to understand whether one hour would be adequate or if it would need to be raised.

To assess skin exposure: Aim the axis of the cone upward to measure any downward radiant energy. If there are any metal or other potentially UV-C-reflective objects in the upper-room irradiated zone (e.g., cameras, light fixtures, TV monitors), aim the detector with FOV safety cone toward each of them to test for reflections. Now, walk over to each wall opposite a UV fixture and aim the axis of the detector cone toward the fixture to capture the direct rays from the fixture at eye level. If above the TWA exposure limit accepted for standing persons, move along that ray toward the fixture until the detector is 1.8 m (about 6 ft) above the floor, unless the fixture is mounted very high above the line of sight, in which case aim the meter below the fixture in a horizontal direction, again from approximately 1.8 m above the floor. (In large rooms, the maximal eye-level measurement may occur at 8 to 10 m from the fixture.) If wall reflectance is of concern (i.e., from high-reflectance paints) or there are many aluminum fixtures on the walls, you would want to measure from each wall position as well as from a center-room position where people could stand.

Step 3: Seated Occupancy Safety. Measure the ocular irradiance at the sitting positions as in an office, restaurant, or waiting room. Perform the same series of measurements as in Step 2, for a standing occupant, but at a height of approximately 1.3 m (about 4 ft 3 in.) above the floor. In these measurements, the cone axis of the meter is aimed horizontally (to measure vertical irradiation). The applicable TWA limit here would normally be a total effective irradiance of 0.1 $\mu\text{W}/\text{cm}^2$ (or 0.2 $\mu\text{W}/\text{cm}^2$ at 254 nm). The values at 222 nm are nearly four times higher.

CAUTION: It is critical that this safety evaluation, as part of the commissioning process, be performed with proper TWA (time-weighted averaging) of the potential exposures in the installation. For example, in addition to the most common cases described above, there could be individuals looking upward, as in a patient ward, in which case the 8-hour limit would apply with the cone axis aimed from the bed level upward to the ceiling and toward the fixtures. As another example, if UV-C fixtures are placed at the front of a classroom, conference room, or lecture hall, the occupants of the space might spend several hours staring nearly directly at the fixtures. The assumption here is that the occupant has his or her eyes open all or most of time relevant time period.

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Process for Change to an ANSI/IES Standard under Continuous Maintenance

This standard is maintained under continuous maintenance procedures, for which IES has an established and documented program for regular publication of addenda or revisions, including procedures for timely, documented, consensus action on requests for change to any part of the standard. Committee consideration will be given to proposed changes by June 30 of any given year for proposed changes received by the IES Director of Standards no later than December 31 of the previous year.

Submittal Format

Proposed changes must be submitted to the IES Director of Standards in the announced published format. However, changes may be accepted in an earlier published format, if the differences are immaterial to the proposed change submittal. If the Director of Standards concludes that a current form must be utilized, the proposer may be given up to 20 additional days to resubmit the proposed changes in the current format.

Specific changes in the text or values are required and must be substantiated. Any change proposals that do not meet these requirements will be returned to the proposer. Supplemental background documents to support changes submitted may be included.

Submission to the Committee Chair

The Director of Standards shall forward proposed changes received on appropriate forms to the committee chair for assigning to committee members (responders) to develop responses to submitters of proposed changes.

Review and Clarification

Responders shall review proposals and should contact the proposer if necessary for clarification.

Response Recommendation

Designated responders shall draft a recommended committee response, including any recommended changes to the standard. The 'responders' recommended responses shall be submitted to the committee chair in electronic form usable by Society Staff, including any recommended change to the standard in response to proposals received.

Options for Committee response are limited to:

- a) Proposed change accepted for public review without modification
- b) Proposed change accepted for public review with modification
- c) Proposed change accepted for further study
- d) Proposed change rejected

The responders shall provide reasons for any recommendation other than option (a) above.

The designated responders shall not recommend option (c) unless the further study can be completed by October 1 of that year, and providing the Committee can then vote for option (a), (b), or (d) no later than November 15 of that year.

Editing

The Committee chair or his or her designee shall edit the draft responses and circulate the edited drafts to the committee for review.

Form for Proposing Change to an ANSI/IES Standard under Continuous Maintenance

NOTE: Use a separate form for each comment. Submit to the Director of Standards, IES, 120 Wall Street, 17th Floor, New York, NY 10005-4001. Email: standards@ies.org. Fax: 212-248-5017.

1. Submitter: _____
Affiliation: _____
Address: _____
City: _____ State: _____ Zip: _____ Country: _____
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I hereby grant the Illuminating Engineering Society (IES) the non-exclusive royalty rights, including non-exclusive rights in copyright, in my proposals. I understand that I acquire no rights in publication of the standard in which my proposals in this, or other analogous, form are used. I hereby attest that I have the authority and am empowered to grant this copyright release.

Submitter's signature: _____ Date: _____

2. Title of publications and year published _____

3. Clause (section), sub-clause or paragraph number; and page number: _____

4. My proposal (check one):

- ☐ Change to read as follows
- ☐ Delete and substitute as follows
- ☐ Add new text as follows
- ☐ Delete without substitution

Use underscore to show material to be added (added) and strikethrough for material to be deleted (~~deleted~~). Use additional pages if needed.

5. Proposed change:

6. Reason and substantiation:

Select as applicable:

- ☐ Additional pages are attached. Number of additional pages: _____
- ☐ Attachments or referenced materials cited in this proposal accompany this proposed change.

Please verify that all attachments and references are relevant, current, and clearly labeled to avoid processing and review delays. Please list your attachments here:



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